



Fundamentals of Optics

Lecturer

Yanjun LIU (刘言军)

E-mail: yjliu@sustech.edu.cn

Room 239, South Tower, College of Engineering
Department of Electrical and Electronic Engineering
Southern University of Science and Technology
Shenzhen, Guangdong, China

Teaching Assistant

Zhenming WANG (王振铭)

E-mail: 12150025@mail.sustech.edu.cn

Room 338, South Tower, College of Engineering

Qing-an TU (涂清安)

E-mail: 12231044@mail.sustech.edu.cn

Room 320, South Tower, College of Engineering

Q&A time: 2-4 pm on Monday

Room 239, South Tower, College of Engineering

Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

CHAPTER 7

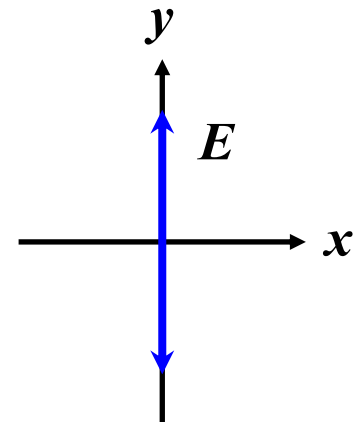
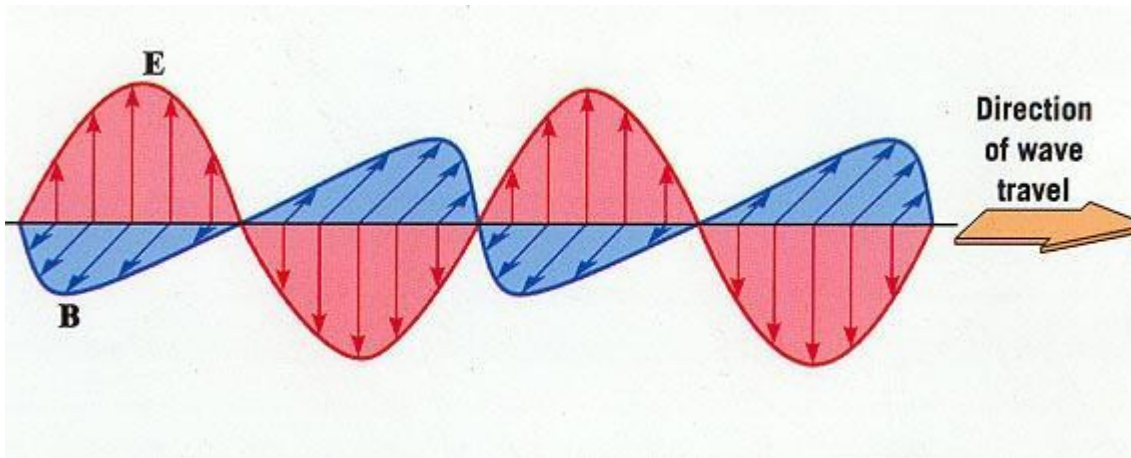
POLARIZATION

7.1 The nature of polarized light

四种实现偏振的方法

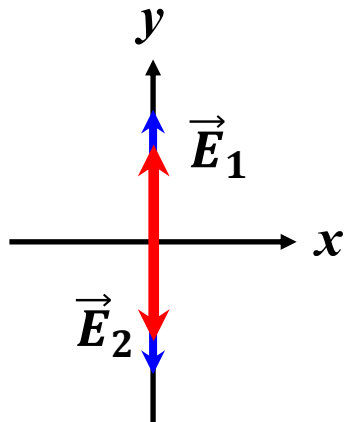
Light: a **transverse electromagnetic wave**.

Linearly polarized or plane-polarized light: the orientation of the electric field is constant, and it resides in **plane-of-vibration**.



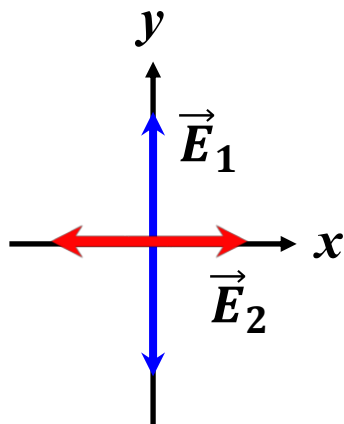
Case: two harmonic, linearly polarized lightwaves of the same frequency, move through the same region of space.

Electrical field vectors are colinear: **superposition**.



Interference, diffraction, beat, ...

Electrical field vectors are perpendicular:



what will happen?

Linear polarization

Two orthogonal optical disturbance:

$$\vec{E}_x(z, t) = \hat{i}E_{0x} \cos(kz - \omega t)$$

$$\vec{E}_y(z, t) = \hat{j}E_{0y} \cos(kz - \omega t + \epsilon)$$

ϵ is the relative phase difference between the waves.

The resultant optical disturbance:

$$\vec{E}(z, t) = \vec{E}_x(z, t) + \vec{E}_y(z, t)$$

Case I: $\epsilon = 2m\pi, m=0, \pm 1, \pm 2, \dots$

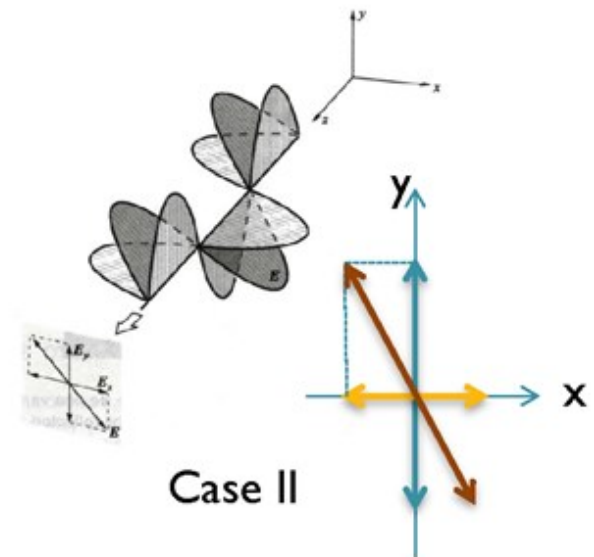
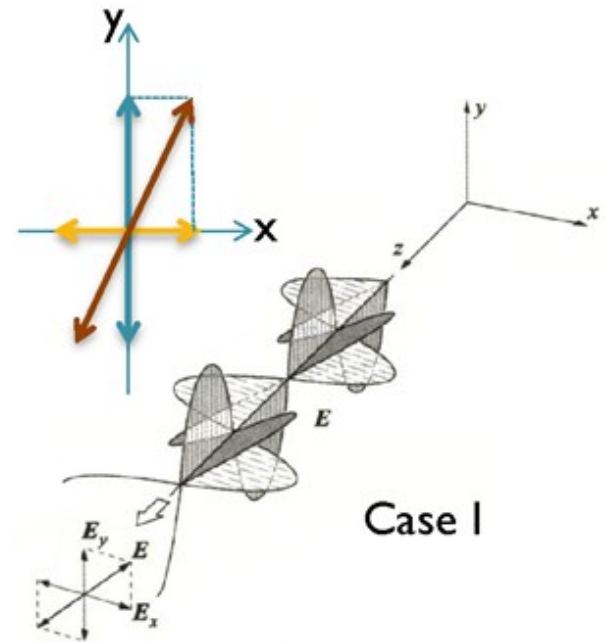
$$\vec{E} = (\hat{i}E_{0x} + \hat{j}E_{0y}) \cos(kz - \omega t)$$

Case II: $\epsilon = (2m+1)\pi, m=0, \pm 1, \pm 2, \dots$

$$\vec{E} = (\hat{i}E_{0x} - \hat{j}E_{0y}) \cos(kz - \omega t)$$

The amplitude is fixed, so it is linearly polarized.

The orientation is tilted.



Circular Polarization

Case III: $\epsilon = -\pi/2 + 2m\pi$, $m=0, \pm 1, \pm 2, \dots$ and $E_{0x} = E_{0y} = E_0$

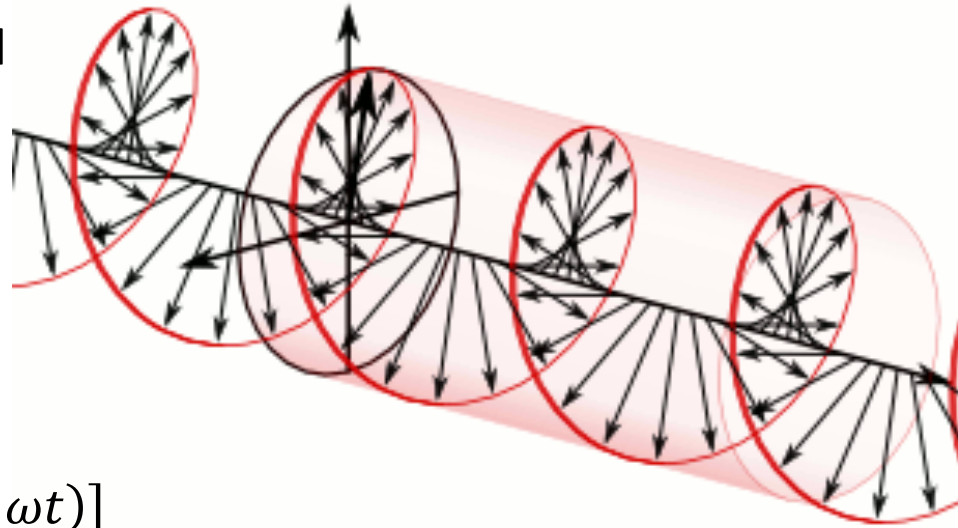
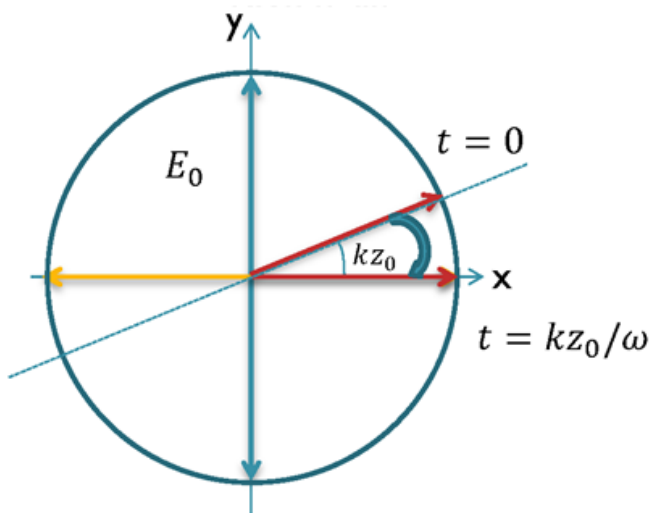
$$\vec{E}_x(z, t) = \hat{i}E_0 \cos(kz - \omega t)$$

$$\vec{E}_y(z, t) = \hat{j}E_0 \sin(kz - \omega t)$$

The consequent wave:

$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)]$$

- ❑ The scalar amplitude is a constant;
- ❑ The direction is time-varying.



For example:

$$t = 0: \vec{E}_x = \hat{i}E_0 \cos kz_0 \text{ and } \vec{E}_y = \hat{j}E_0 \sin kz_0$$

$$t = kz_0/\omega: \vec{E}_x = \hat{i}E_0 \text{ and } \vec{E}_y = 0$$

The resultant electric-field vector $\mathbf{E}$ is rotating clockwise, as seen by an observer toward moving direction of waves, **right-circularly polarized light** (right-circular light)

Case IV: $\epsilon = \pi/2 + 2m\pi$, where $m = 0, \pm 1, \pm 2, \dots$ and $E_{0x} = E_{0y} = E_0$

The consequent wave:

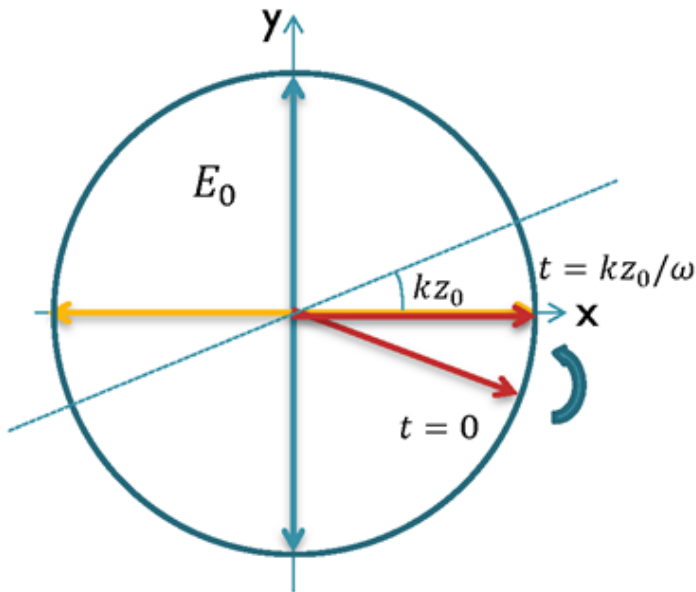
$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)]$$

The resultant electric-field vector **E** rotates counterclockwise,
left-circularly polarized.

For example:

$$t = 0: \vec{E}_x = \hat{i}E_0 \cos kz_0 \text{ and } \vec{E}_y = -\hat{j}E_0 \sin kz_0$$

$$t = kz_0/\omega: \vec{E}_x = \hat{i}E_0 \text{ and } \vec{E}_y = 0$$



❖ A linearly polarized wave can be synthesized from two oppositely polarized circular waves of equal amplitude.

$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)]$$

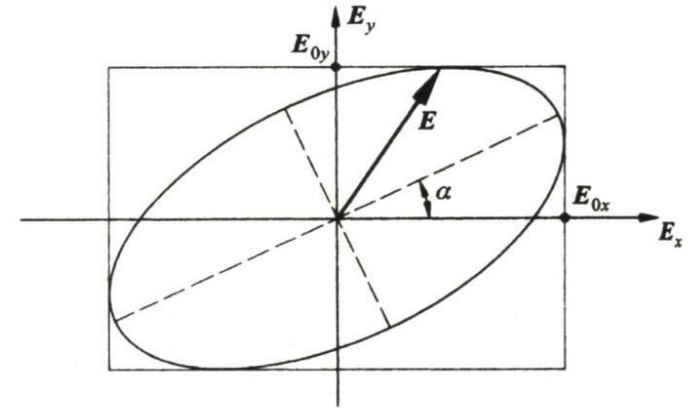
$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)]$$

$$\vec{E} = 2E_0\hat{i} \cos(kz - \omega t)$$

Elliptical polarization

Case V: $E_{0x} \neq E_{0y}$ and ϵ is random

The resultant electric-field vector $\mathbf{E}$ will rotate, and change its magnitude.



$$E_x = E_{0x} \cos(kz - \omega t)$$

$$E_y = E_{0y} \cos(kz - \omega t + \epsilon)$$

Elliptically polarized light

$$\left(\frac{E_y}{E_{0y}}\right)^2 + \left(\frac{E_x}{E_{0x}}\right)^2 - 2\left(\frac{E_x}{E_{0x}}\right)\left(\frac{E_y}{E_{0y}}\right)\cos\epsilon = \sin^2\epsilon$$

This is the equation of an ellipse making an angle α with the (E_x, E_y) -coordinate system

$$\tan 2\alpha = \frac{2E_{0x}E_{0y}\cos\epsilon}{E_{0x}^2 - E_{0y}^2}$$

$$\alpha = 0 \text{ or } \epsilon = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \dots \quad \left(\frac{E_y}{E_{0y}}\right)^2 + \left(\frac{E_x}{E_{0x}}\right)^2 = 1$$

$$E_{0x} = E_{0y} = E_0$$

$$E_x^2 + E_y^2 = E_0^2$$

$$\epsilon = 2m\pi, m = 0, 1, 2, \dots$$

$$E_y = \frac{E_{0y}}{E_{0x}} E_x$$

$$\epsilon = (2m + 1)\pi, m = 0, 1, 2, \dots$$

$$E_y = -\frac{E_{0y}}{E_{0x}} E_x$$

**Elliptically
polarized light**

**Circularly
polarized light**

**Linearly
polarized light**

State of Polarization

Linearly polarized light: $\mathcal{P}$ -state

Right-or left- circular light: $\mathcal{R}$ or $\mathcal{L}$ -state

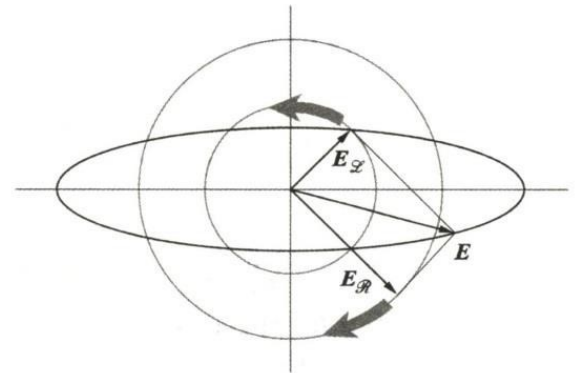
Elliptical polarization: $\mathcal{E}$ -state

$\mathcal{P}$ -state can be represented as a superposition of $\mathcal{R}$ and $\mathcal{L}$ -state.

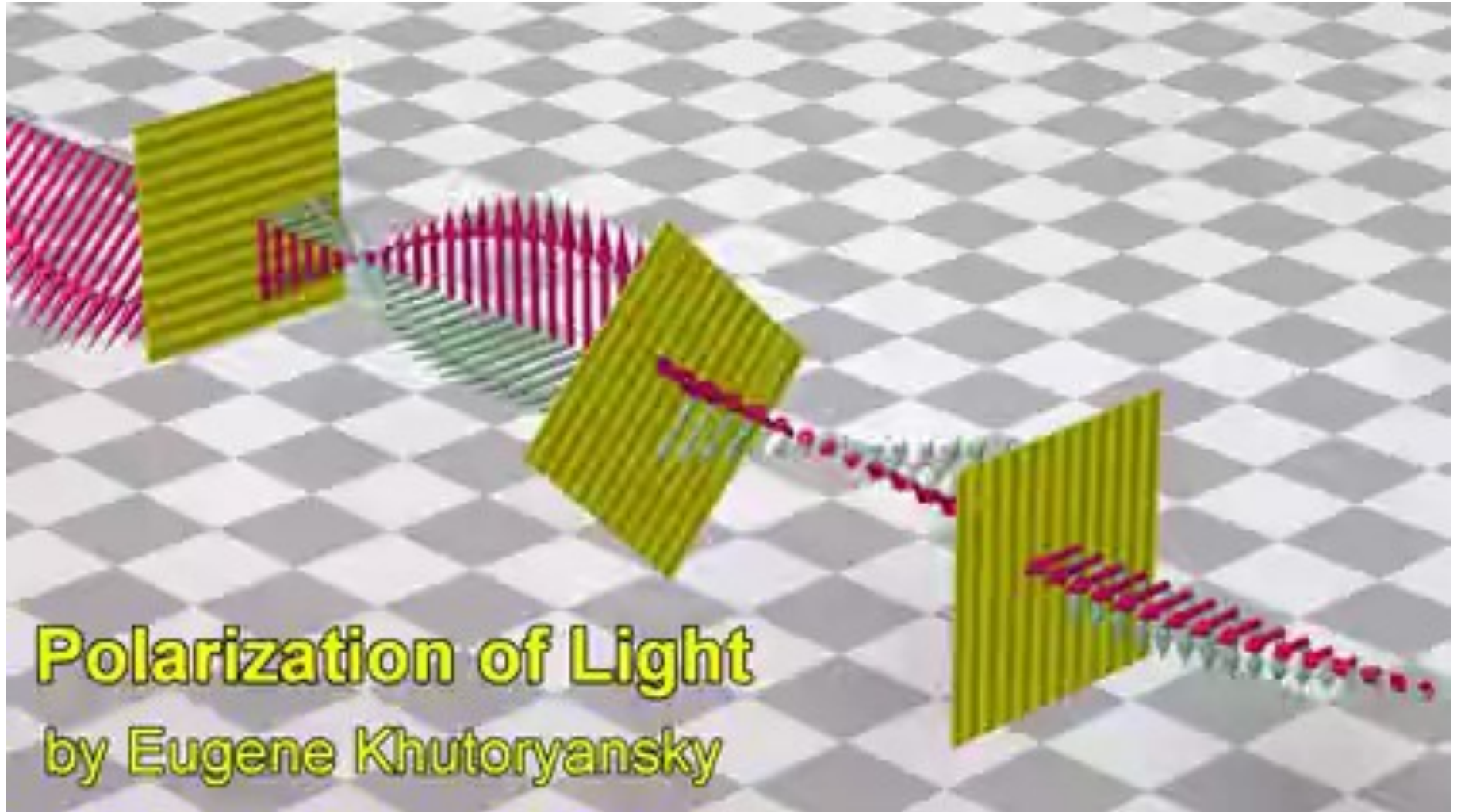
$$\begin{aligned}\vec{E} &= E_0[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)] \\ &+ E_0[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)] = 2E_0\hat{i} \cos(kz - \omega t)\end{aligned}$$

$\mathcal{E}$ -state also can be represented as superposition of $\mathcal{R}$ and $\mathcal{L}$ -state with different amplitudes

$$\begin{aligned}\vec{E} &= E_{01}[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)] \\ &+ E_{02}[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)] \\ &= \hat{i}(E_{01} + E_{02}) \cos(kz - \omega t) \\ &+ \hat{j}(E_{01} - E_{02}) \sin(kz - \omega t)\end{aligned}$$

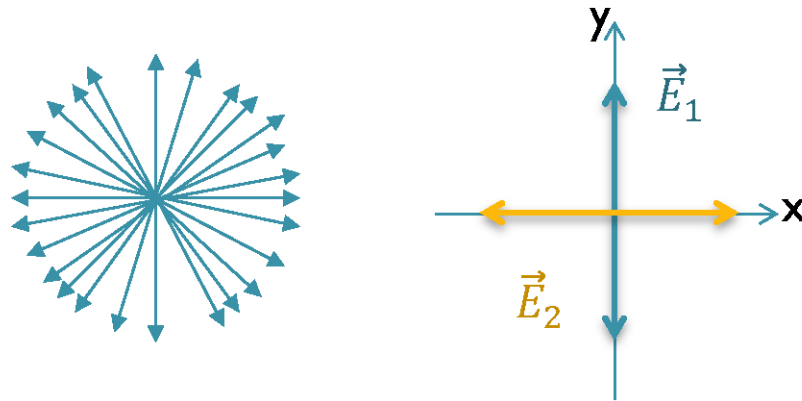


A video introduction about the polarization of light

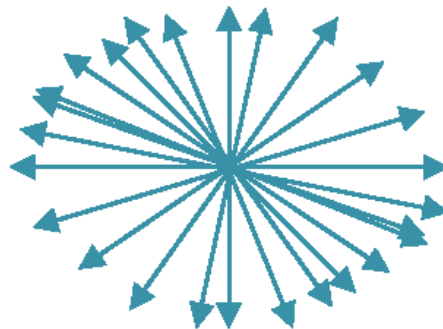


Natural Light

Natural light: the electromagnetic waves emitted from any common source of light are polarized randomly. That is, the electric field changes direction randomly.



Partially polarized waves: light is generally neither completely polarized nor completely unpolarized. (superposition of specific amounts of natural and polarized light).



Angular momentum and the photon picture

Classical picture: electromagnetic wave impinging on an object can impart both energy and linear momentum to that body.

- If the incident plane wave is circularly polarized, the resulting motion is again circular.
- The E-field drives the electron with an angular frequency ω equal to the frequency of the electromagnetic wave.

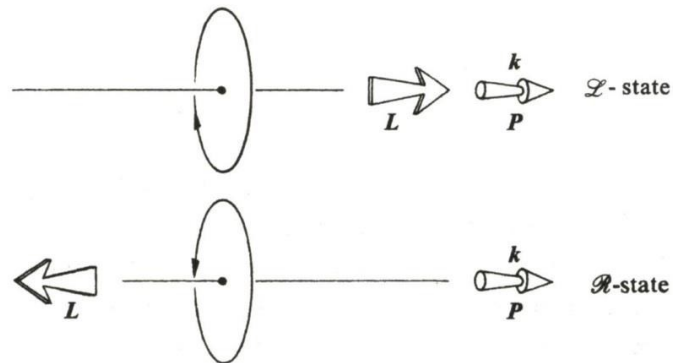
The power (the energy transferred per unit time)

$$\frac{d\xi}{dt} = \omega \mathbf{\Gamma} = \omega \frac{dL}{dt}$$

$\mathbf{\Gamma}$: the torque;
 L : the angular momentum.

$$L = \frac{\xi}{\omega}$$

A charge that absorbs a quantity of energy from the incident circular wave will simultaneously absorb an amount of angular momentum.



Quantum-mechanical description:

An electromagnetic wave transfers energy in quantized packets (photons):

$$\xi = h\nu \quad \xi = \hbar\omega \quad (\hbar \equiv \frac{h}{2\pi})$$

The intrinsic or spin angular momentum of a photon is either $-\hbar$ (right-handedness) or $+\hbar$ (left-handedness).

*The spin angular momentum of a photon is completely independent of its energy.

When a charged particle emits or absorbs electromagnetic radiation, it will undergo a change of $\pm\hbar$ in its angular momentum.

The linearly polarized light beam will impart no angular momentum to a target.

7.2 Polarizers

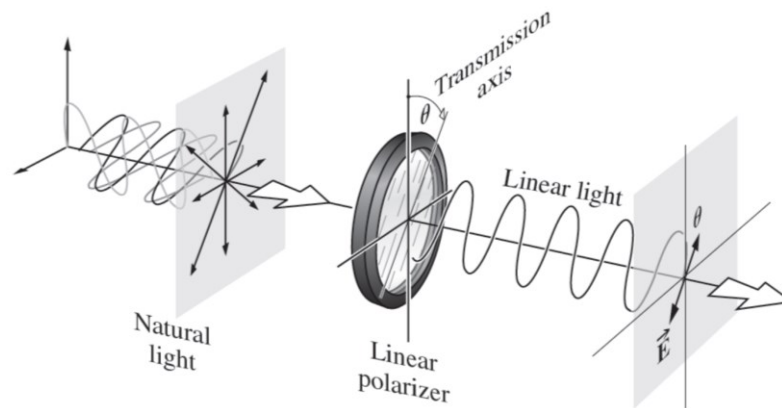
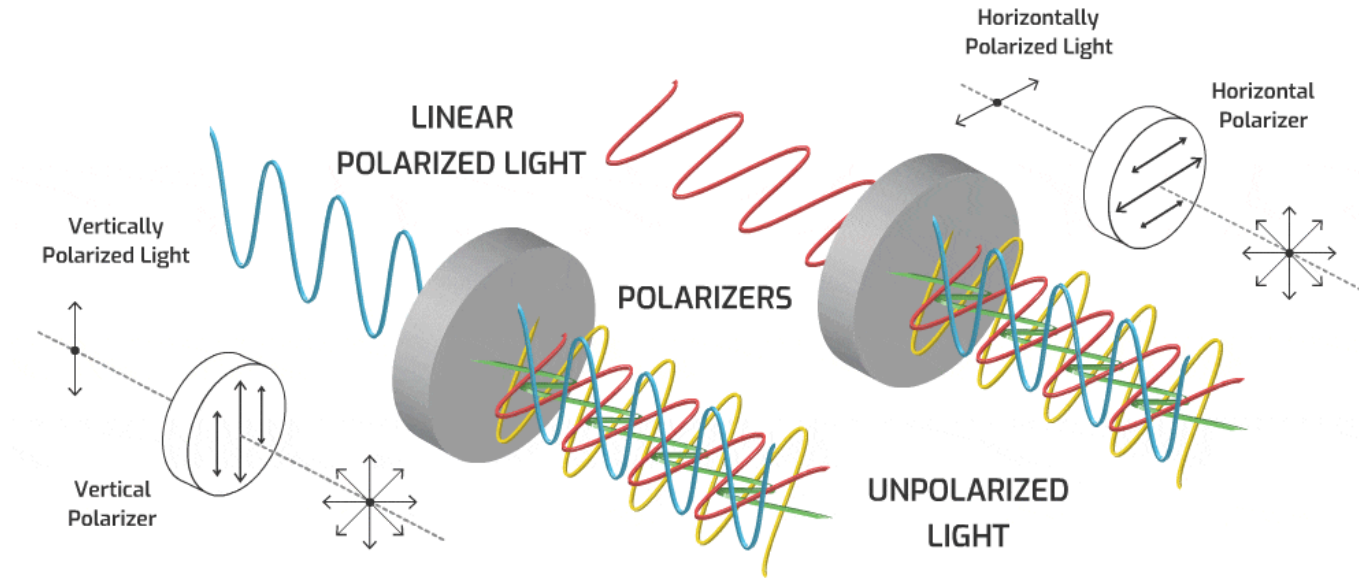
Polarizer: an optical device whose input is natural light and whose output is some form of polarized light.

Linear polarizer; circular polarizer; elliptical polarizer and partial polarizer.

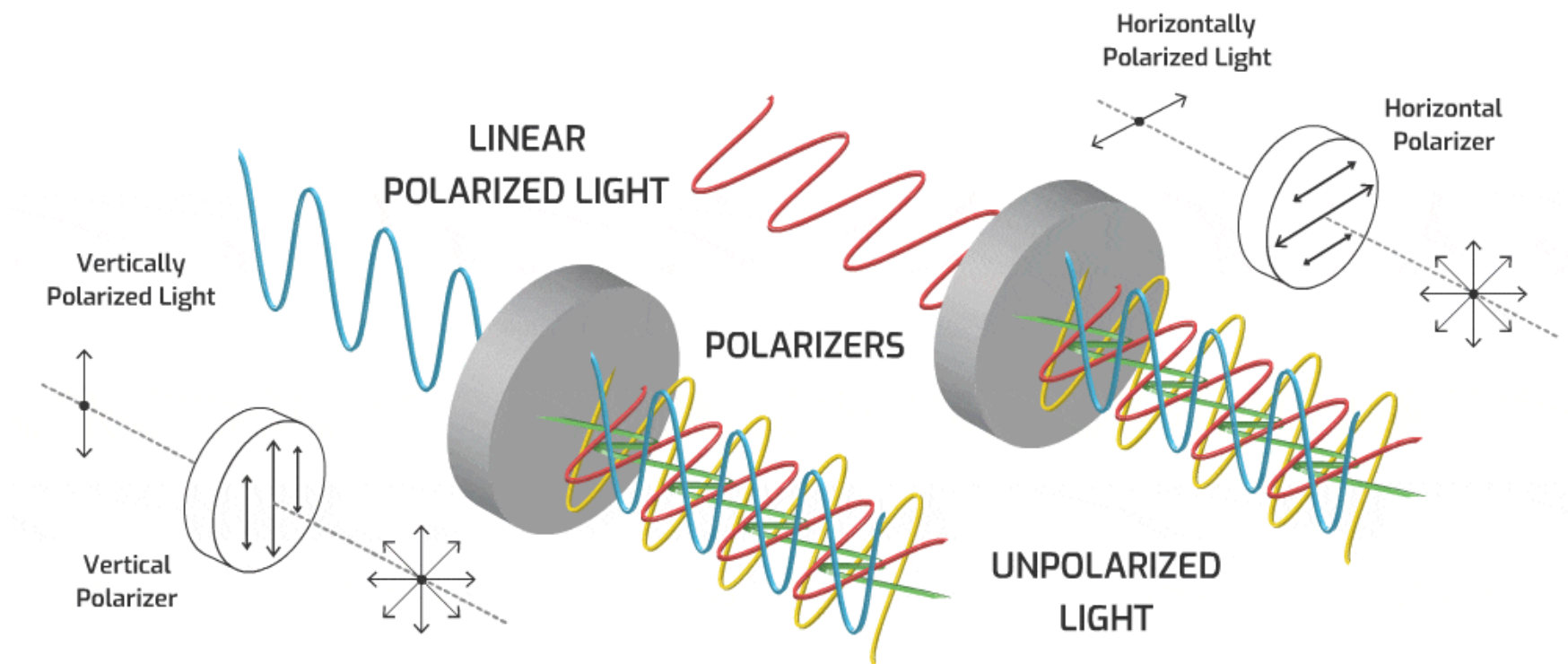
Fundamental physical mechanisms of polarizer:

- ☐ Dichroism (二向色性)/ selective absorption;
- ☐ Reflection;
- ☐ Scattering;
- ☐ Birefringence (双折射)/ double refraction

Optical transmission of a linear polarizer



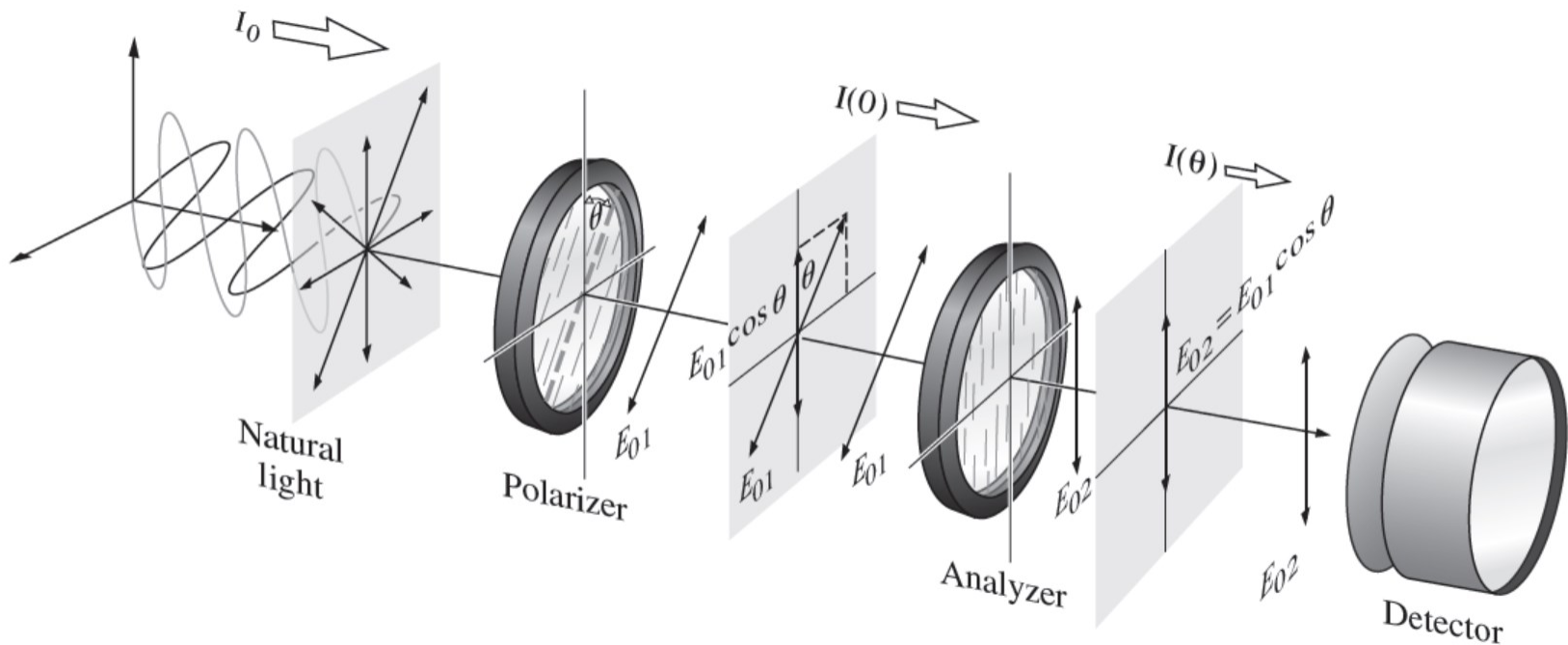
We can only measure intensity (irradiance) $I = \frac{c\epsilon_0}{2} E^2$



Malus's law (马吕斯定律)

Intensity of transmitted light through a linear polarizer

$$I(\theta) = \frac{c\epsilon_0}{2} E_{01}^2 \cos^2 \theta = I(0) \cos^2 \theta$$

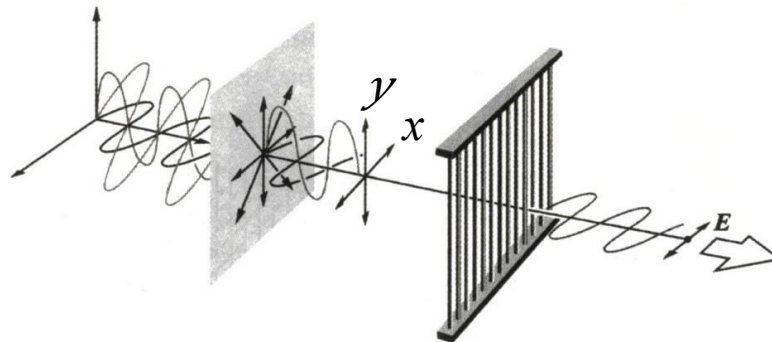


7.3 Dichroism

Dichroism: the selective absorption of one of the two orthogonal $\mathcal{P}$ -state components of an incident beam.

The wire-grid polarizer

The simplest device is a grid of parallel conducting wires.

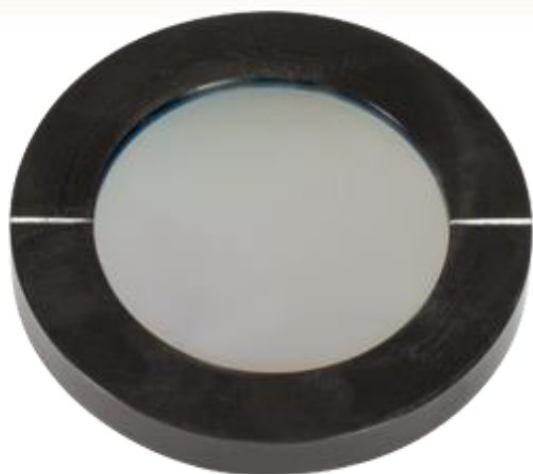


How it works:

- ❑ The y -component of the field $\rightarrow$ drive conduction electrons along the length of each wire $\rightarrow$ collide with lattice atoms (joule heat) $\rightarrow$ energy is transferred from the field to grid.
- ❑ Electrons accelerating along the y -axis $\rightarrow$ radiate in the forward and backward direction $\rightarrow$ the incident wave cancel the wave reradiated in the forward direction $\rightarrow$ no transmission of the y -component of field
- ❑ The electrons are not free to move very far in x -component $\rightarrow$ no absorption
- ❑ The transmission axis of the grid is perpendicular to the wires

Commercial product

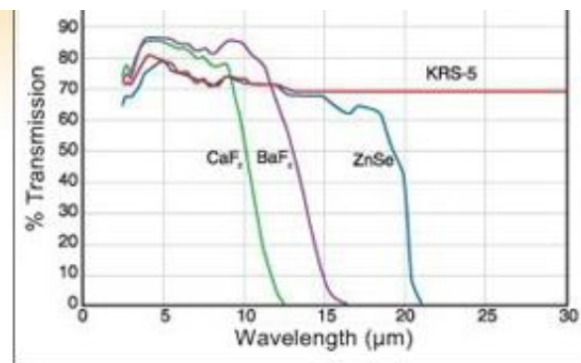
► BaF₂, CaF₂, KRS-5, and ZnSe Substrate Options



WP50H-B
(Ø50 mm)



WP25H-Z
(Ø25 mm)



Dichroic Crystals Certain materials are inherently dichroic because of an anisotropy in their respective crystalline structure.

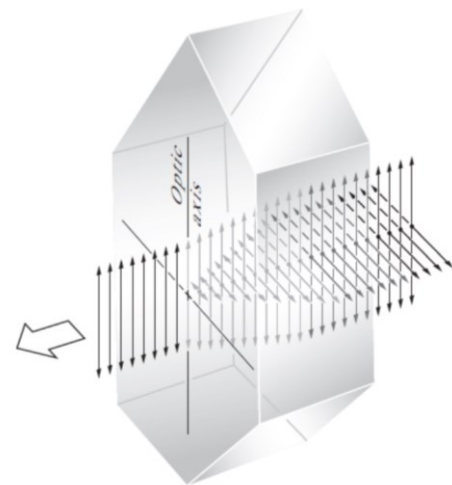
For example: tourmaline (电气石)

- ❑ Principal or optical axis determined by its atomic configuration: electric-field component perpendicular to the principal axis is strongly absorbed.
- ❑ The crystal's principal axis becomes the polarizer's transmission axis.

Polaroid (人造偏振片):

J-sheet: very small needle-shaped herapathite (碘硫酸奎宁) were aligned nearly parallel by means of magnetic or electric fields or narrow slits.

H-sheet: it contains aligned molecules [polyvinyl alcohol (聚乙烯乙醇) with iodine] analogue of the wire grid.



Homework

□ 8.1, 8.3, 8.21

7.4 Birefringence

Many crystalline substances are optically anisotropic. It can be described using a mechanical model depicting a negatively charged shell bound to a positive nucleus by pairs of springs having different stiffness.

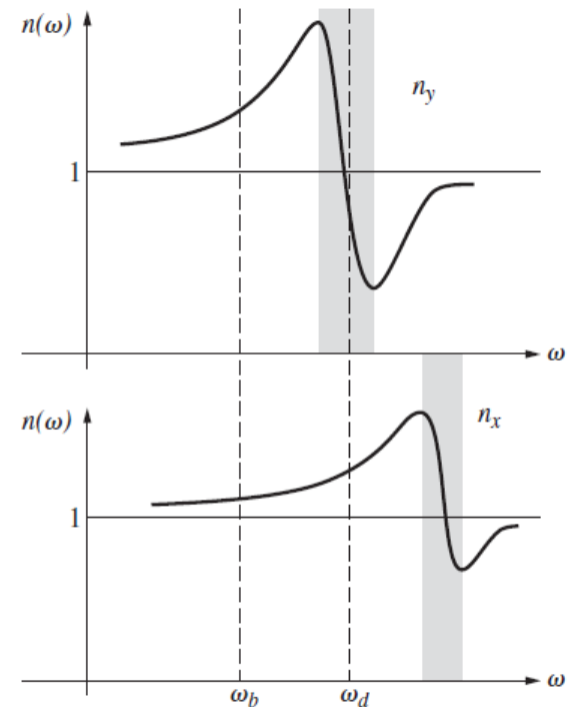
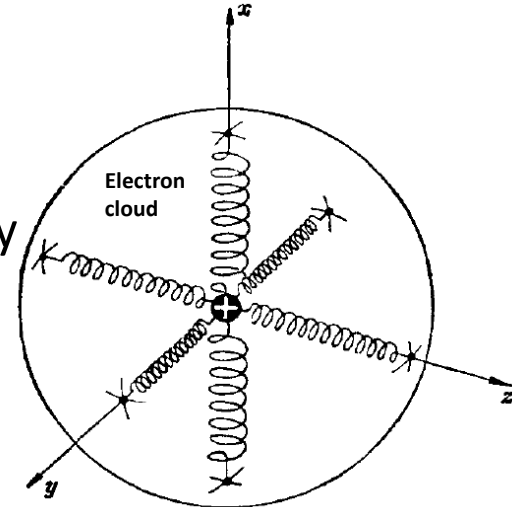
Model: The refractive index n

$$n^2(\omega) = 1 + \frac{Nq^2}{\epsilon_0 m} \left(\frac{1}{\omega_0^2 - \omega^2} \right)$$

ω and ω_0 : the angular frequency of the driving force and the natural angular frequency of the electric dipole, respectively.

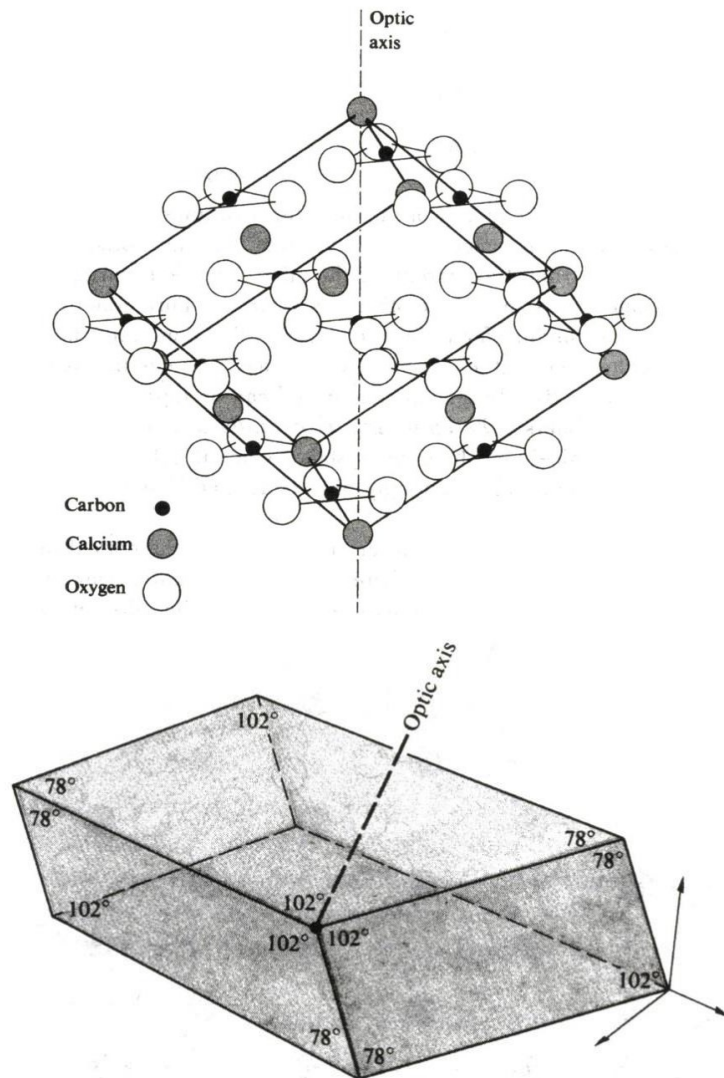
In the vicinity of ω_d , one of the orthogonal $\mathcal{P}$ -state is absorbed and the other passes. It is **dichroic**.

If the stiffness of the spring in x direction has one value while has another value along any direction perpendicular to x -direction, the material has two different refractive indices (In the vicinity of ω_d) and is said to be **birefringent**. The x -direction is called the **optic axis**.



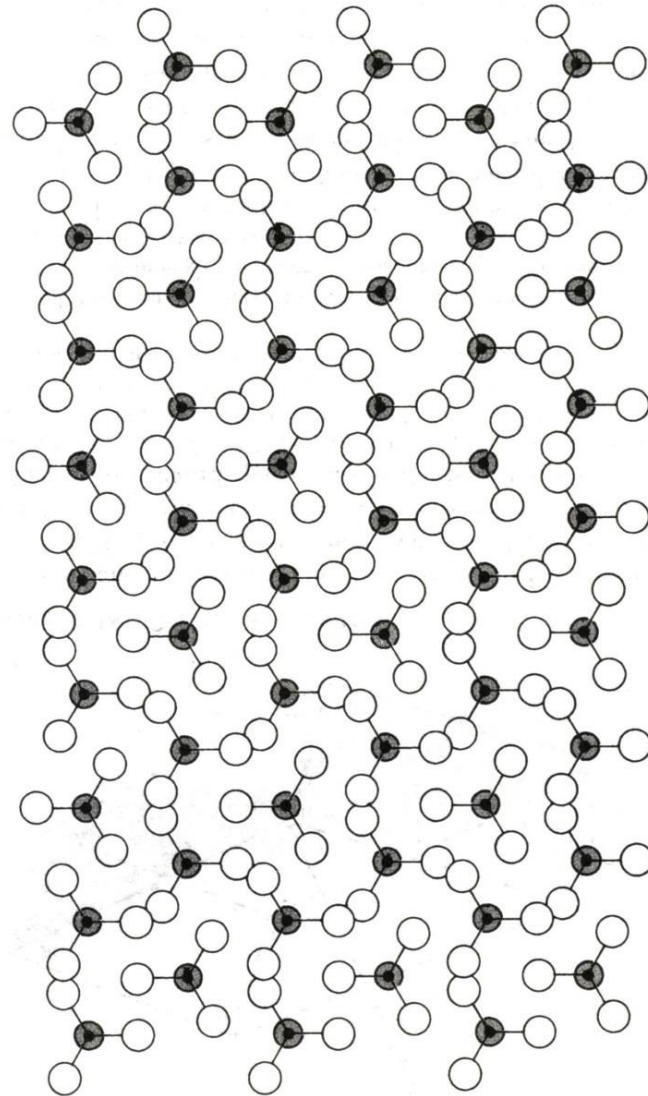
Calcite (方解石)

A typical birefringent crystal (CaCO_3)



Cleavage form

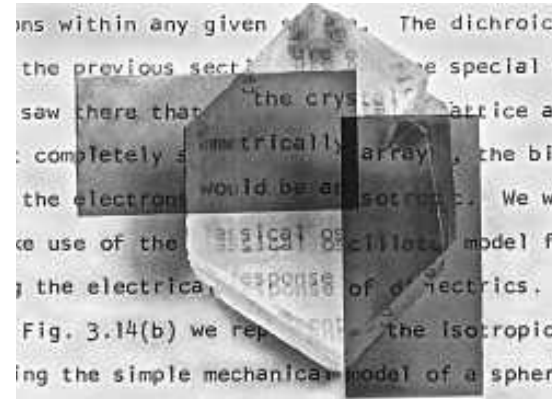
Rhombohedron (斜方六面体)



Ordinary rays (o-rays)
Extraordinary rays (e-rays)

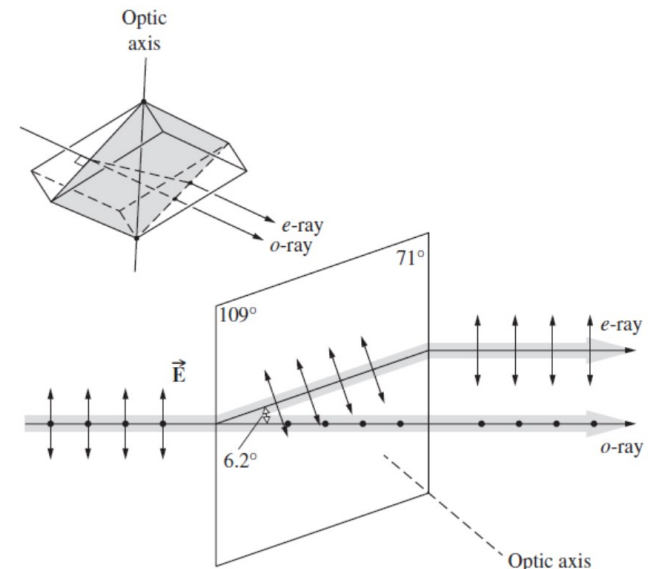


The ordinary and extraordinary images are linearly polarized, and two $\mathcal{P}$ -state are orthogonal.

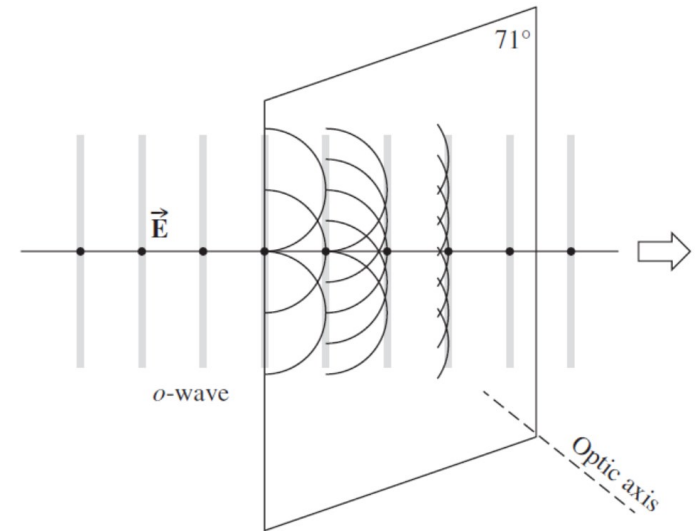


Principal planes (主平面): any number of planes is drawn through the rhomb so as to contain the optic axis.

Principal section (主截面): if the principal plane is normal to a pair of opposite surfaces of the cleavage form.

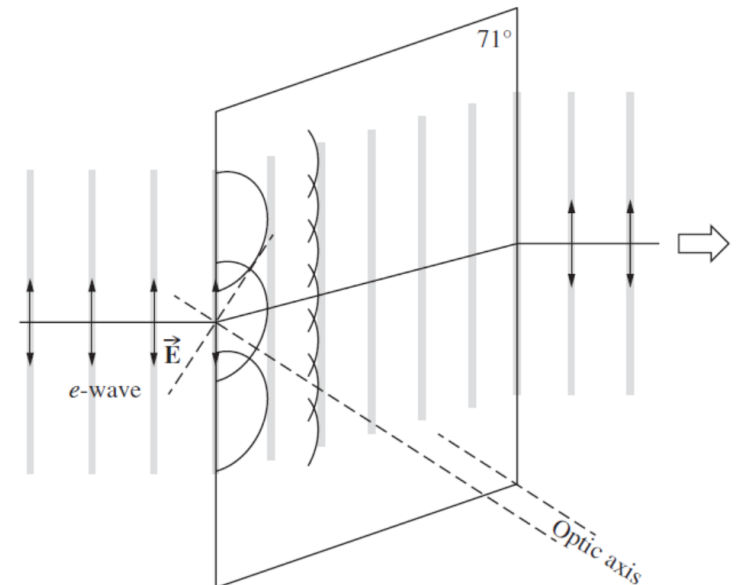
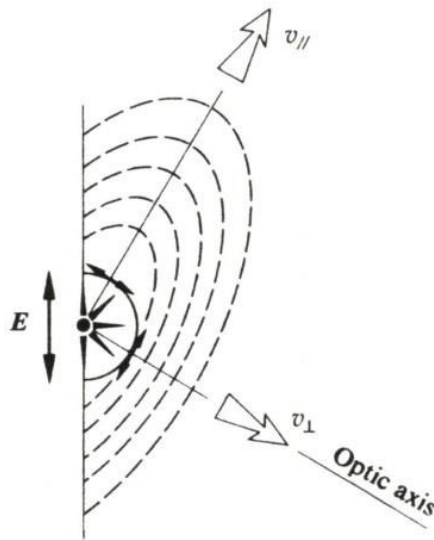


The incident plane wave is linearly polarized and perpendicular to the optic axis.
 The velocity $v_{\perp}$ in all direction is the same.



E-field is parallel to the principal section:

- ❑ a component normal to the optic axis $v_{\perp}$;
- ❑ a component parallel to it $v_{\parallel}$. But $v_{\perp} \neq v_{\parallel}$



Birefringent crystals

The optic axis: a direction about which the atoms are arranged symmetrically.

Uniaxial: crystals with only one such direction.

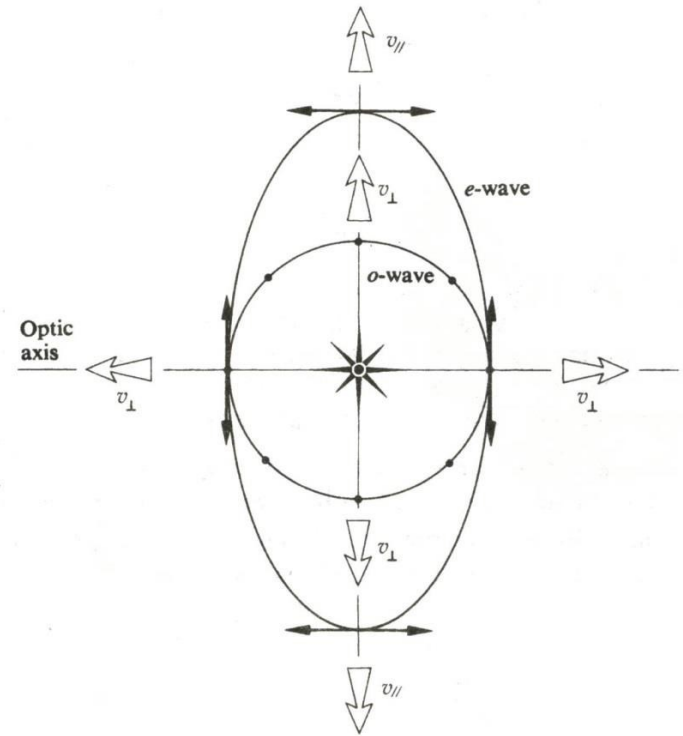
Ordinary rays (o-rays): The E-field of the o-wave is everywhere normal to the optic axis, so it moves at a speed $v_{\perp}$ in all directions.

Extraordinary rays (e-rays):

- ❑ The E-field of e-wave has a speed $v_{\perp}$ only in the direction of the optic axis
- ❑ E-field parallel to the optic axis: speed $v_{\parallel}$

Uniaxial materials have two principle indices of refraction:

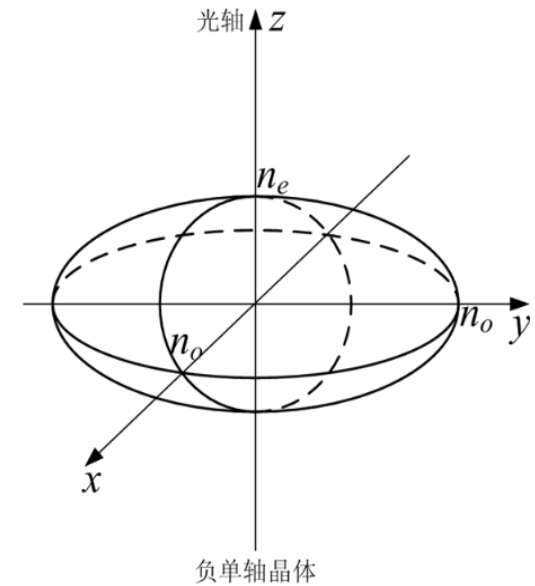
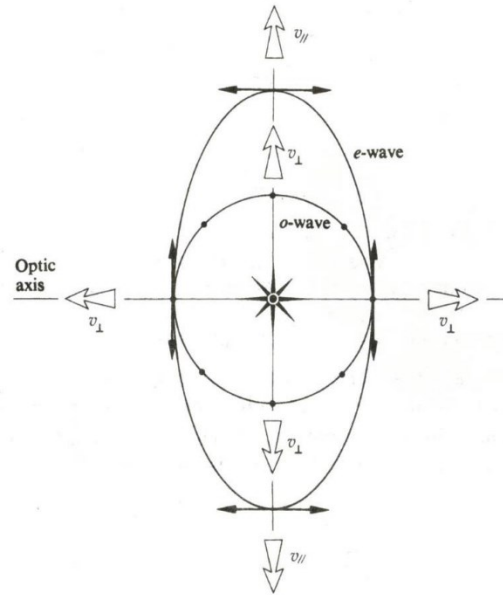
$$n_o \equiv \frac{c}{v_{\perp}} \qquad n_e \equiv \frac{c}{v_{\parallel}}$$



Birefringence: the difference $\Delta n = n_e - n_o$

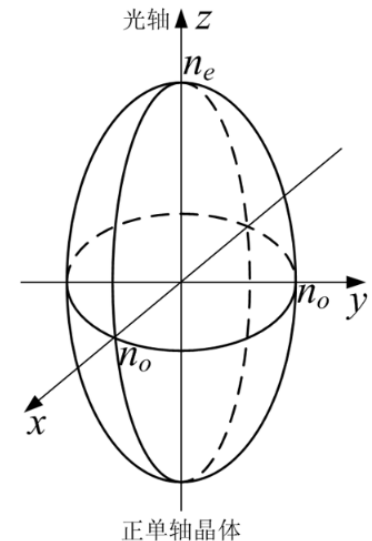
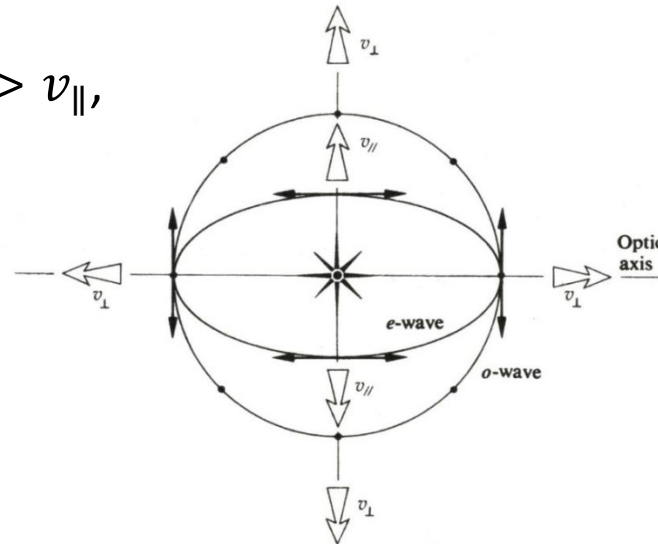
Calcite $v_{\parallel} > v_{\perp}$, $\Delta n = -0.172$,
negative uniaxial.

Calcite $n_o = 1.658$; $n_e = 1.486$



Other crystals (quartz and ice): $v_{\perp} > v_{\parallel}$,
 Δn is positive, positive uniaxial.

Quartz $n_o = 1.544$; $n_e = 1.553$



Birefringent polarizers

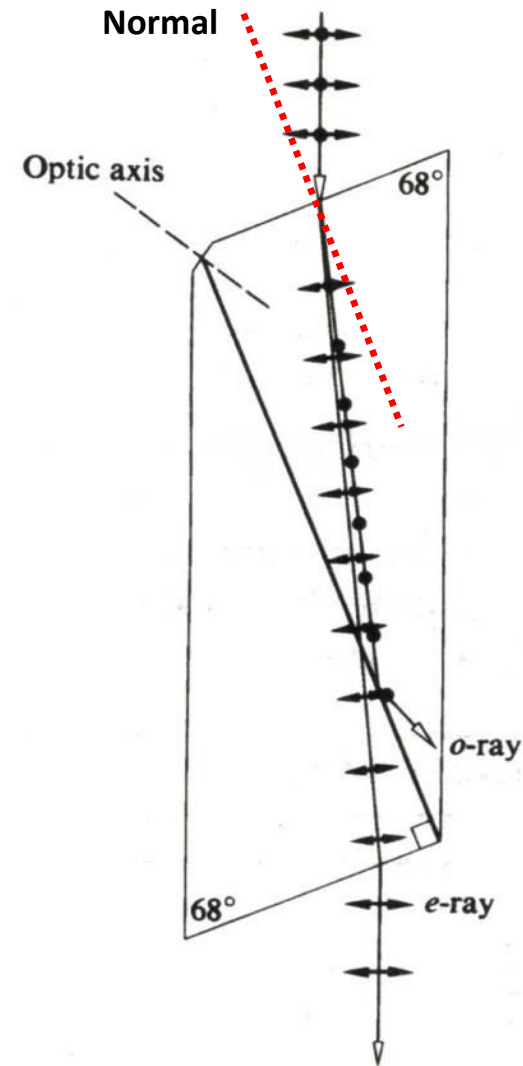
Nicol prism:

The device is made by first grinding and polishing the ends of a suitable long, narrow calcite rhombohedron.

Calcite $n_o = 1.658$; $n_e = 1.486$; $n_o > n_e$; $\theta_{to} < \theta_{te}$

The two pieces are polished and cemented back together with Canada balsam (加拿大树胶) with an index of 1.55, which is almost in the midway between n_e and n_o .

The incident beam is refracted and separated after entering the prism. On the interface, o-ray is totally internally reflected and the e-ray passes.



Glan-Foucault polarizer (Glan-Air polarizer): calcite (5000nm-230nm)

The incoming ray strikes the surface normally, including two components (parallel and perpendicular to the optic axis)

In the first section: two rays traverse without any deviation.

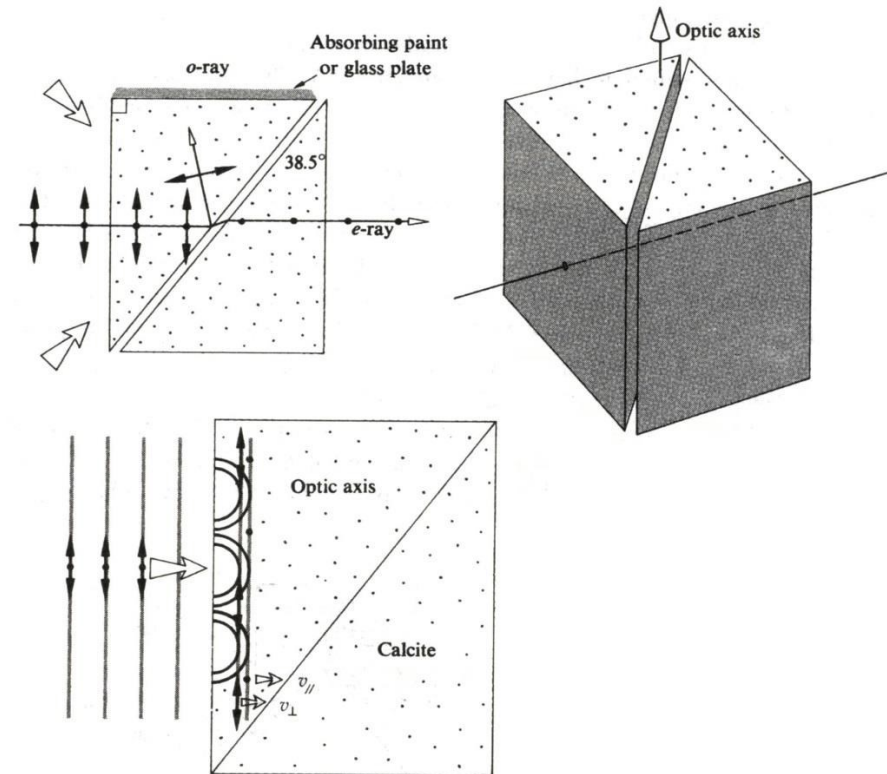
In the calcite-air interference: set

$n_e < \frac{1}{\sin \theta} < n_o$, o-ray is totally internally reflected; e-ray passes.

Glan-Thompson polarizer:

Two prisms are cemented together and interface angle is changed appropriately.

- ❑ Advantage: High field of view (30°), for Glan-Foucault polarizer, 10° .
- ❑ Disadvantage: can not handle high power levels of lasers

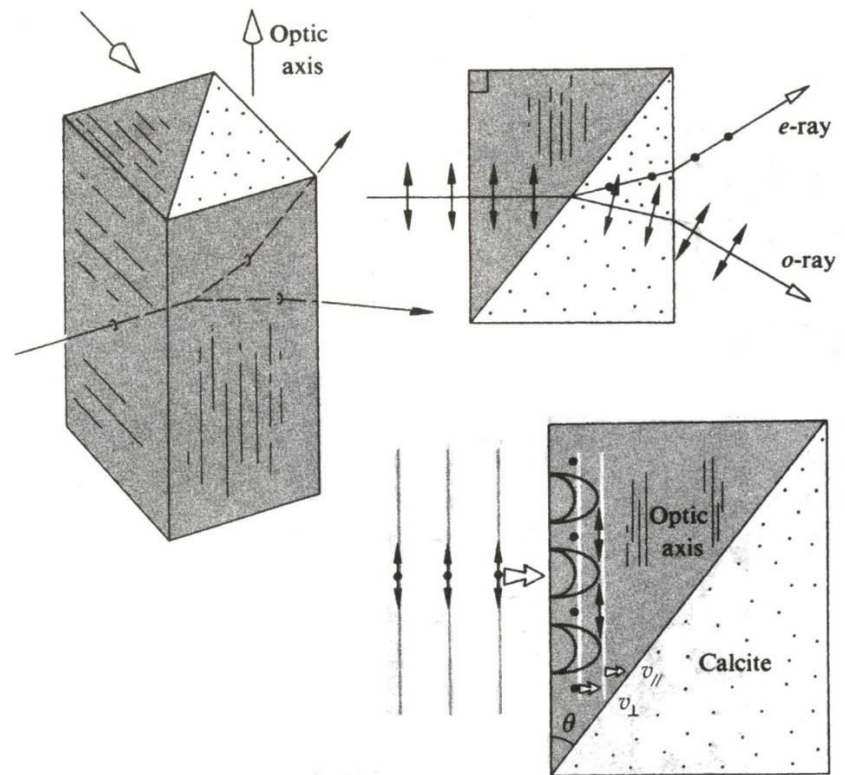


Wollaston prism:

A polarizing beamsplitter, (calcite or quartz).

In calcite: $n_e < n_o$

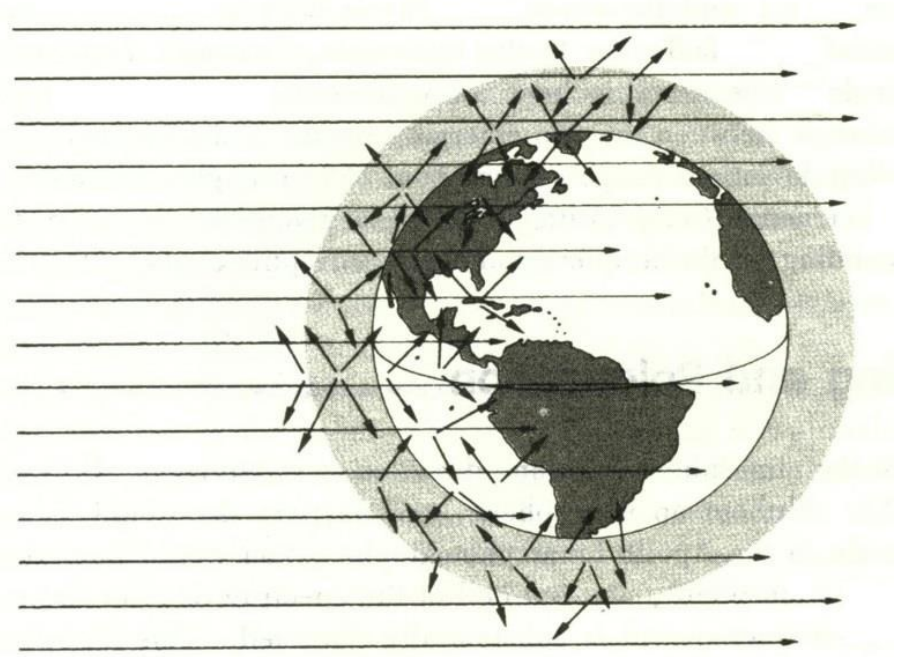
The o-ray, whose field is initially perpendicular to the optic axis, becomes an e-ray, which is bent away from the normal to the interface.



Homework

□ 8.40

7.5 Scattering and Polarization

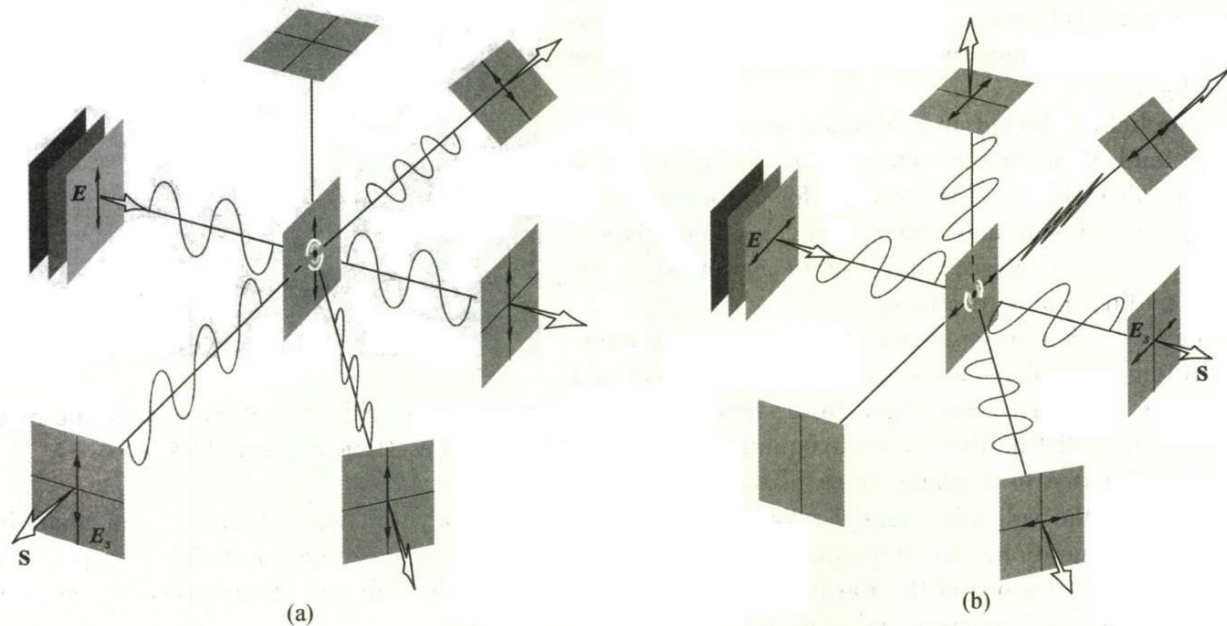


Sunlight streaming into the atmosphere from one direction is scattered in all directions by the air molecules.

7.5.1 Polarization by Scattering

A linearly polarized plane wave incidents on an air molecule;

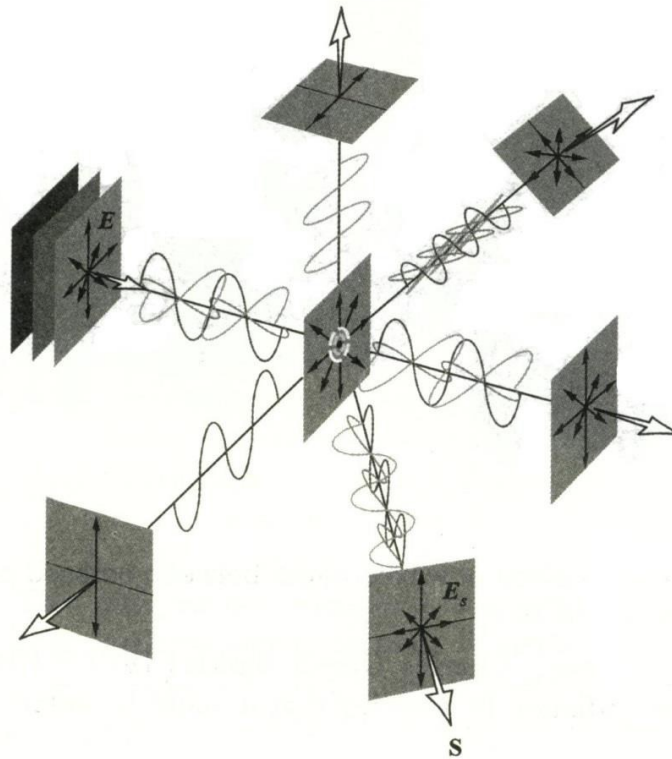
- ❑ The vibrations induced in the atom are parallel to the E-field of the incoming lightwave and perpendicular to the propagation direction.
- ❑ There is no radiation in the direction of its axis.



Scattering of polarized light by a molecule.

As to unpolarized light:

- ❑ The scattered light in the forward direction is completely unpolarized;
- ❑ Off that axis it is partially polarized;
- ❑ When the direction of observation is normal to the primary beam, the light is completely linearly polarized.



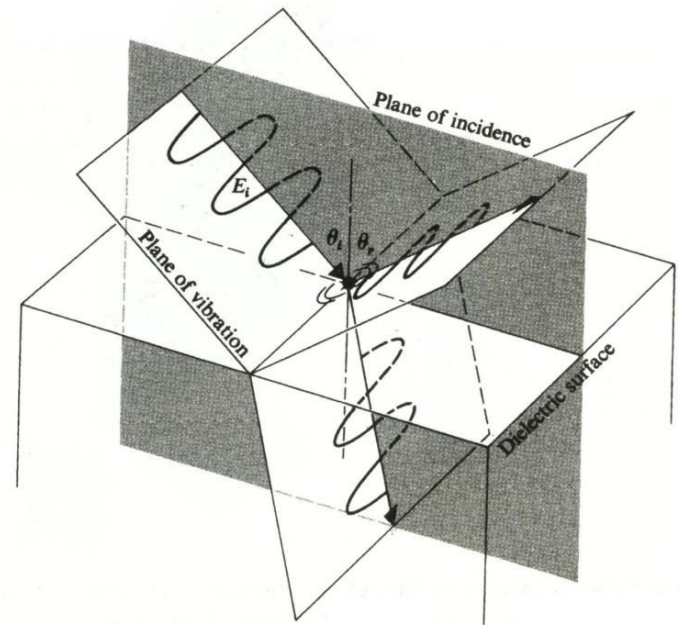
Scattering of unpolarized light by a molecule.

7.6 Polarization by Reflection

The Electron Oscillator Model:

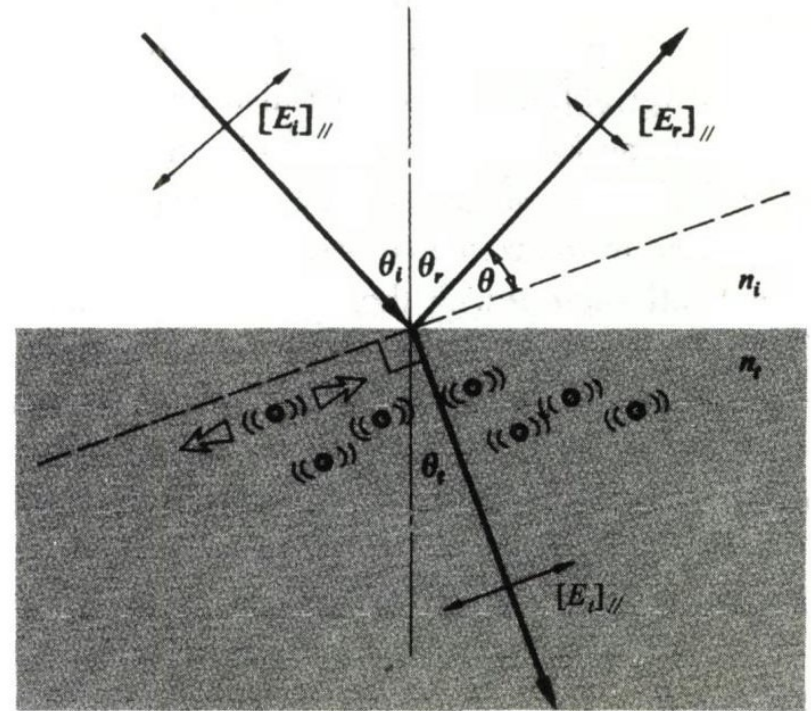
The E-field is perpendicular to the plane of incidence

- ❑ The wave is refracted at the interface, and enter the medium at some transmission angle.
- ❑ Its electric field drives the bound electrons, and they in turn reradiate.
- ❑ A portion of that reemitted energy appears in the form of a reflected wave. And both the reflected and refracted waves must also be in P-states normal to the incident plane.



The E-field is in the incident plane

- ❑ The electron-oscillators near the surface will vibrate under the influence of the refracted wave.
- ❑ The reflected ray direction makes a small angle with the dipole axis and the flux density of reflected wave is relatively low.
- ❑ When $\theta=0$, the reflected wave would vanish entirely.



An incoming unpolarized wave(two incoherent orthogonal P-state): only the component polarized normal to the incident plane will be reflected for the **polarization angle or Brewster's angle**.

When $\theta_p + \theta_t = 90^\circ$,

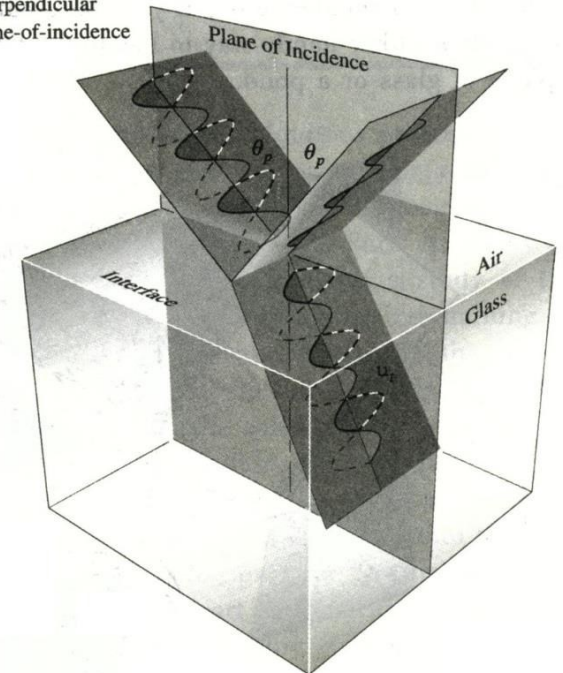
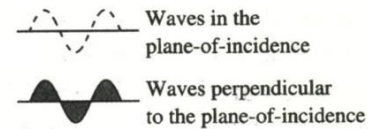
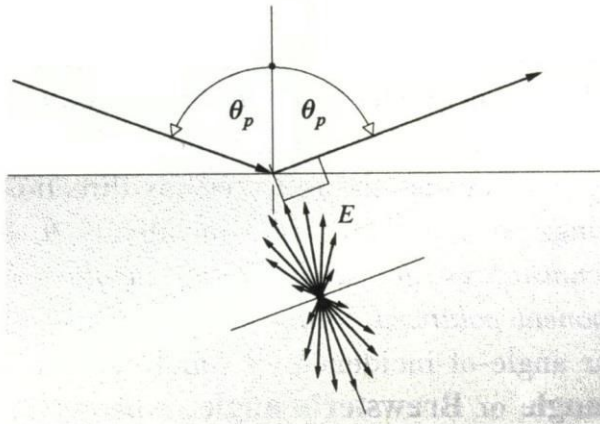
From Snell's Law: $n_i \sin \theta_p = n_t \sin \theta_t$

We have: $\tan \theta_p = n_t/n_i$

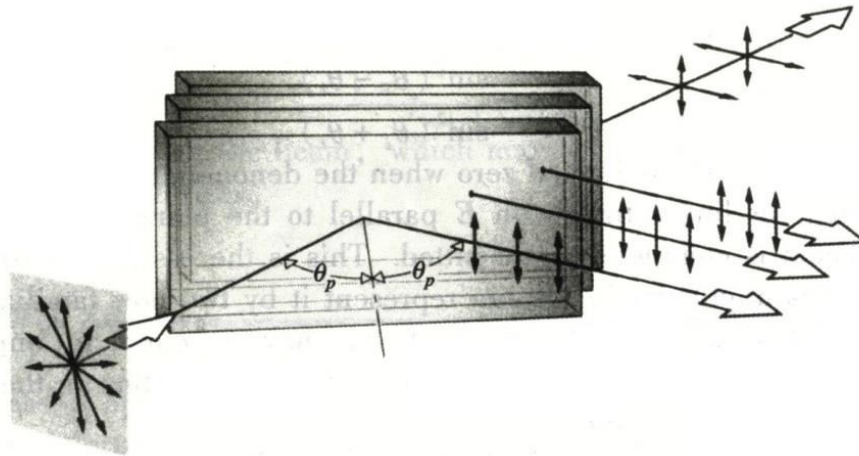
Example:

For air/glass ($n_t \approx 1.5$), the polarization angle is 56° ,

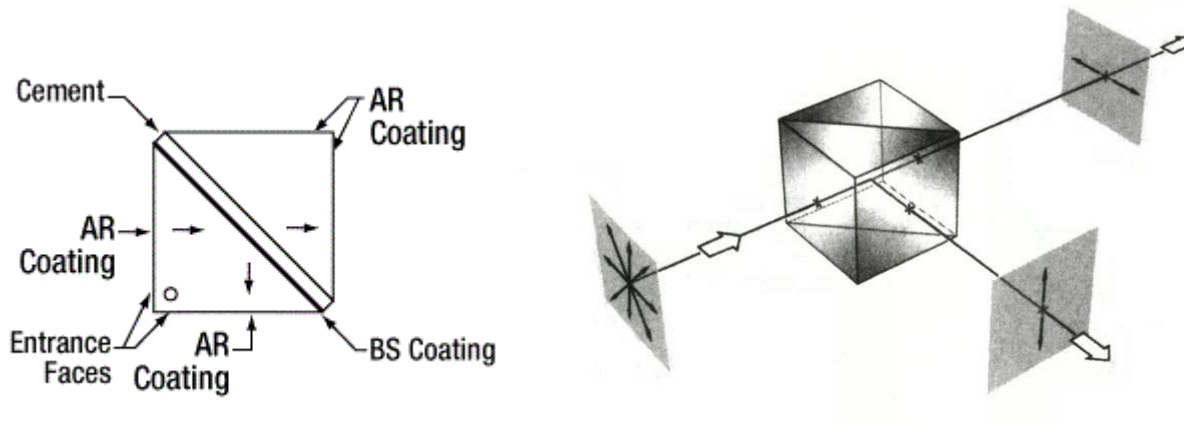
For air/pond ($n_t \approx 1.33$), the polarization angle is 53° .



The pile-of-plates polarizer



A polarizing cube contains a multilayer dielectric thin film structure on its diagonal face.



7.6.1 An Application of the Fresnel Equations

For linear light having its E-field parallel to the plane-of-incidence:

Amplitude reflection coefficient: $r_{\parallel} \equiv [E_{0r}/E_{0i}]_{\parallel}$

For E-field normal to the incident plane:

Amplitude reflection coefficient: $r_{\perp} \equiv [E_{0r}/E_{0i}]_{\perp}$

Reflectance: $R_{\parallel} = r_{\parallel}^2 = [E_{0r}/E_{0i}]_{\parallel}^2$ and $R_{\perp} = r_{\perp}^2 = [E_{0r}/E_{0i}]_{\perp}^2$

Squaring the appropriate Fresnel Equations yields:

$$R_{\parallel} = \frac{\tan^2(\theta_i - \theta_t)}{\tan^2(\theta_i + \theta_t)}$$

$$R_{\perp} = \frac{\sin^2(\theta_i - \theta_t)}{\sin^2(\theta_i + \theta_t)}$$

When $\theta_i + \theta_t = 90^\circ$ (Brewster's angle), namely the denominator is infinite, $R_{\parallel}$ is indeed zero, $R_{\perp}$ can never be zero.

The incoming light is unpolarized ($I_{i\parallel} = I_{i\perp} = I_i/2$)

It is represented by two orthogonal, incoherent, equal-amplitude $\mathcal{P}$ -state.

$$I_{r\parallel} = \frac{I_{r\parallel} I_i}{2I_{i\parallel}} = R_{\parallel} I_i / 2$$

$$I_{r\perp} = \frac{I_{r\perp} I_i}{2I_{i\perp}} = R_{\perp} I_i / 2$$

The reflectance in natural light:

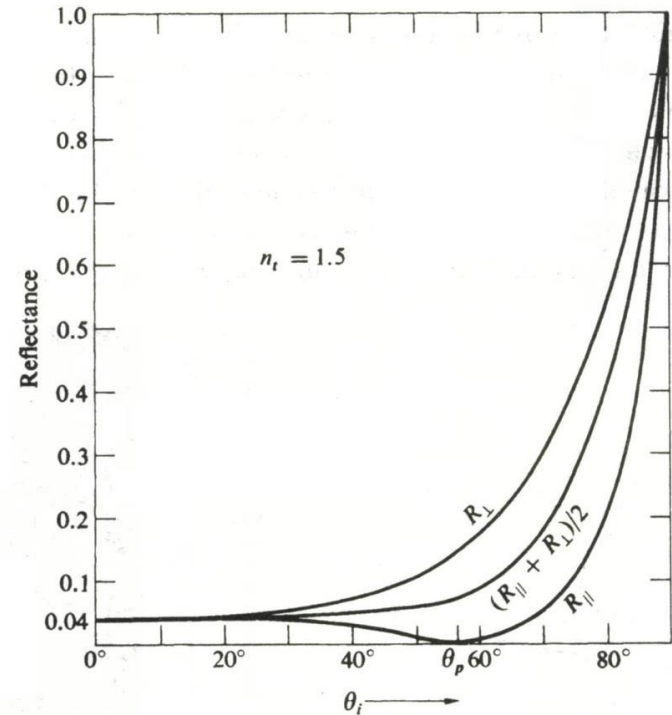
$$R = \frac{I_r}{I_i} = \frac{I_{r\parallel} + I_{r\perp}}{I_i} = \frac{1}{2} (R_{\parallel} + R_{\perp})$$

Note:

- ❑ When $\theta_i = \theta_p$, 7.5% of the incoming light is reflected;
- ❑ When $\theta_i \neq \theta_p$, both the transmitted and reflected waves are partially polarized.

Degree of polarization (偏振度) V : $V = \frac{I_p}{I_p + I_u}$

When using a linear polarizer to check a partially polarized light $V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$

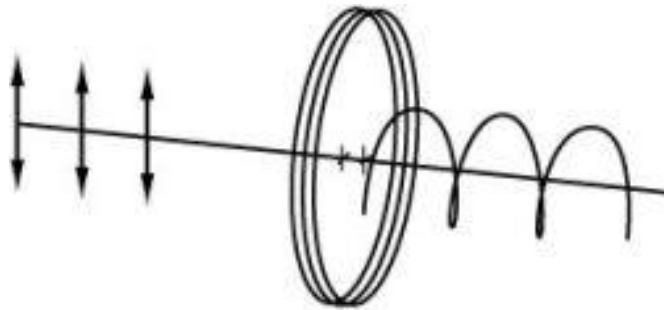


7.7 Retarders

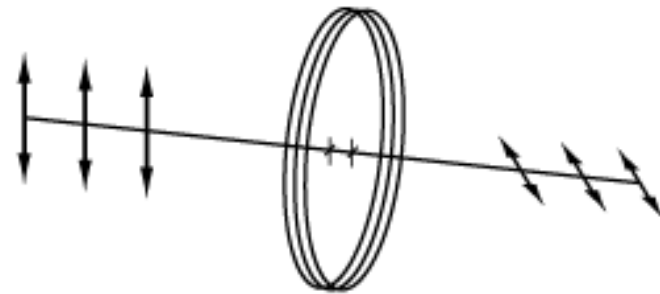
Retarders: the devices changing the polarization of an incident wave.

Principle: one of the two constituent coherent $\mathcal{P}$ -states is caused to have its phase lag behind that of the other by a predetermined amount.

Example:



Quarter-Wave Plates Change Linearly Polarized Light into Circularly Polarized Light.



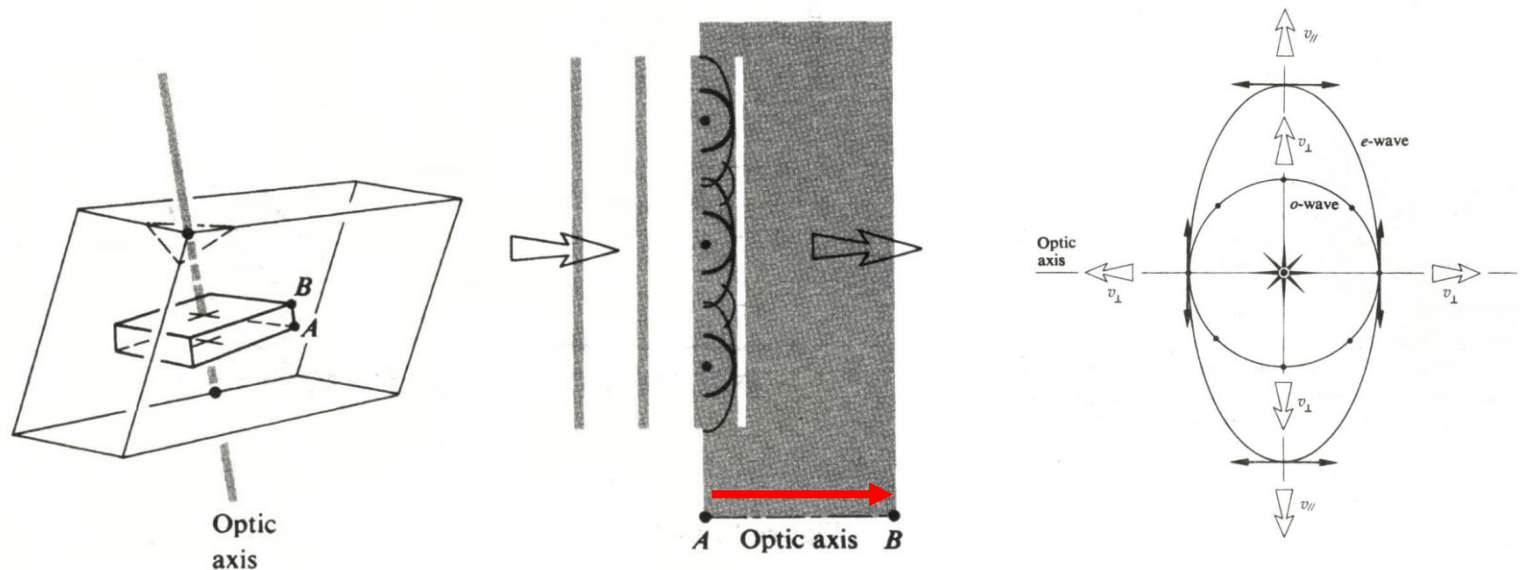
Half-Wave Plates Rotate the Polarization Plane of Linearly Polarized Light.

7.7.1 Wave Plates and Rhombs

I) The optic axis of calcite crystal is normal to both the front and back surfaces.

A normally incident plane wave:

- ❑ Only E-field perpendicular to the optic axis exists;
- ❑ The secondary spherical and ellipsoidal wavelets will be tangent to each other in the direction of the optic axis;
- ❑ The o- and e-waves will be coincident and a single undeflected plane wave will pass through the crystal. **No relative phase shift and no double images.**



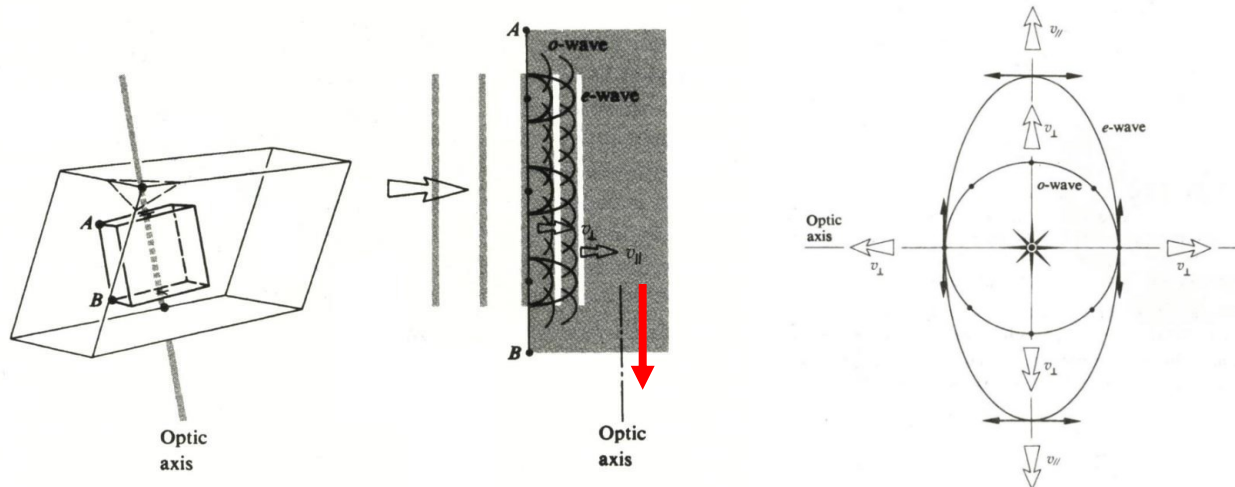
II) The direction of optic axis is parallel to the front and back surfaces.

- ❑ The E-field of an incident monochromatic plane wave has components parallel and perpendicular to the optic axis;
- ❑ Since $v_{\parallel} > v_{\perp}$, $n_o > n_e$, the e-wave will move across the specimen more rapidly than the o-wave.
- ❑ The relative optical path length [$\Lambda = d|n_o - n_e|$] and the phase difference [$\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$] are different.

$$\vec{E}_o(z, t) = \hat{i}E_{0o} \cos(kz - \omega t)$$

$$\vec{E}_e(z, t) = \hat{j}E_{0e} \cos(kz - \omega t + \Delta\varphi)$$

The state of polarization of the emergent light evidently depends on the amplitudes of the incoming orthogonal field components and of course on $\Delta\varphi$



The Full-Wave Plate:

The phase difference:
$$\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$$

The Full-Wave Plate: $\Delta\varphi = 2\pi$, the relative retardation is one wavelength.

Note:

- $|n_o - n_e|$ changes little over the optical range;
- $\Delta\varphi \propto 1/\lambda_0$: a full-wave plate can function only in the manner discussed for a particular wavelength.

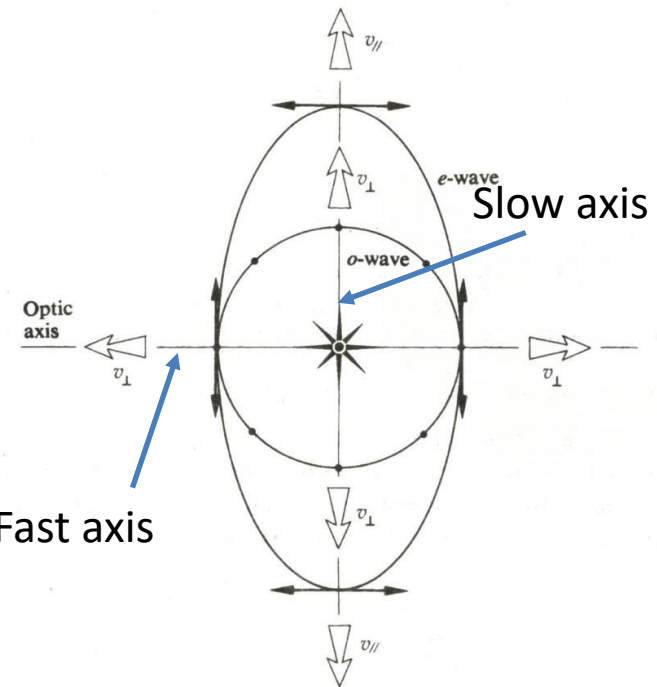
Calcite $v_{\parallel} > v_{\perp}$, $n_e < n_o$, $\Delta n = -0.172$, negative uniaxial.

For negative uniaxial crystal, when the $\vec{E}$ -field is parallel to the optic axis, $v_{\parallel} > v_{\perp}$

Fast axis: the direction of optic axis in a negative uniaxial retarder is referred to as the fast axis.

Slow axis: the direction perpendicular to optic axis.

For positive uniaxial negative uniaxial:
slow axis corresponds to the optic axis.



The Half-Wave Plate

The phase difference: $\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$

The Half-Wave Plate: $\Delta\varphi = \pi$, the relative retardation is $1/2$ wavelength.

Note:

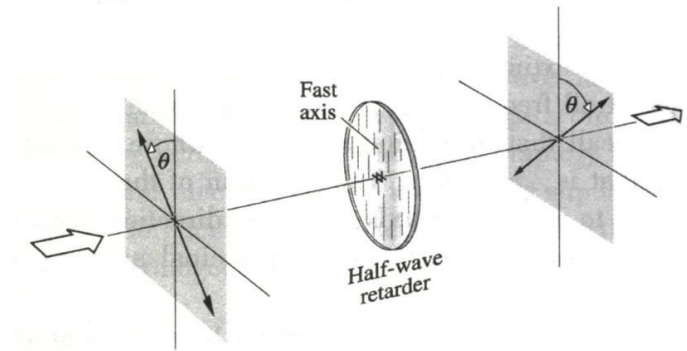
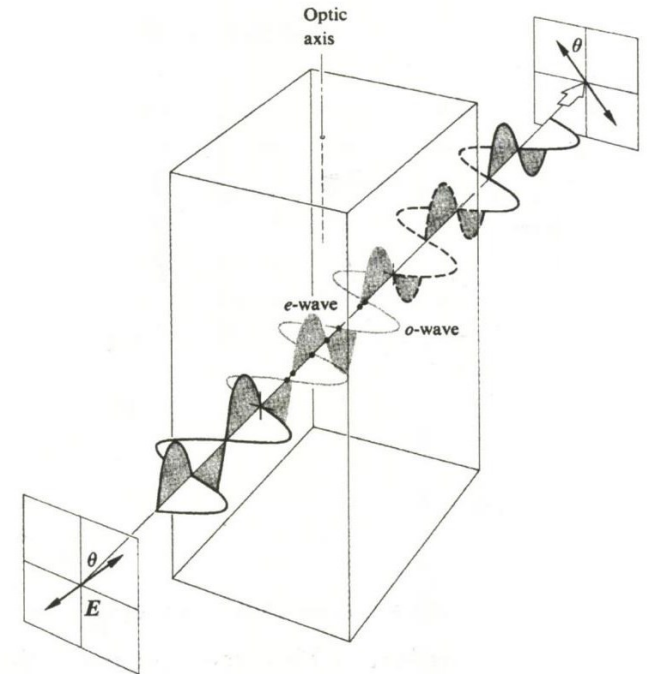
- ❑ As to elliptical light, it will have the same function.
- ❑ As to handedness of circular and elliptical light, it will invert it.

The thickness of the materials must satisfy:

$$d(|n_o - n_e|) = (2m + 1) \lambda_0 / 2$$

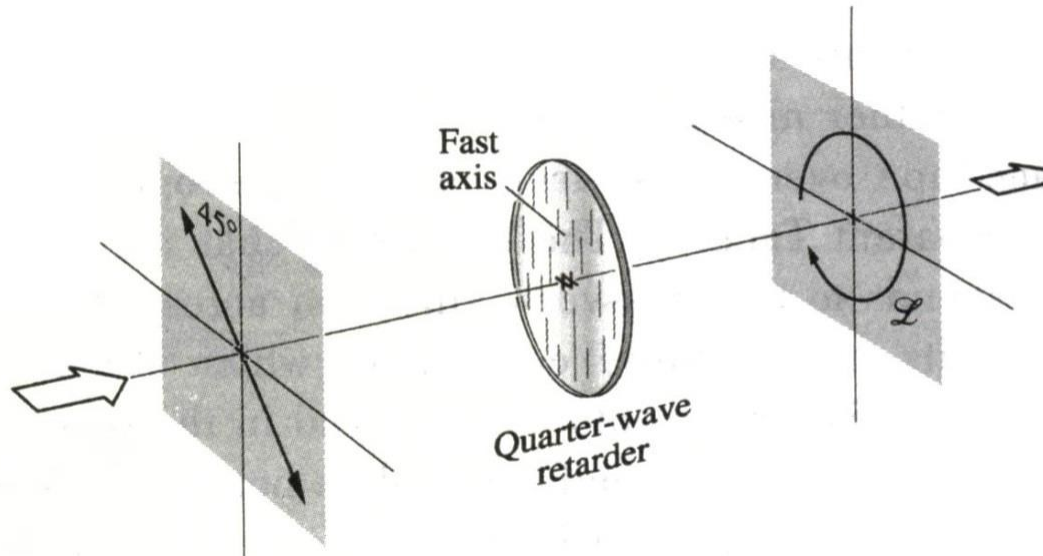
$$m = 0, 1, 2, \dots$$

Materials: Mica, crystalline quartz, single crystal magnesium fluoride (for IR), and cadmium sulfide (for IR)



The Quarter-Wave Plate

The Quarter-Wave Plate: $\Delta\varphi=\pi/2$ between the constituent orthogonal o- and e-components of a wave.



When linear light at 45° to either principle axis incident on a quarter-wave plate, its o- and e-components have equal amplitude, a 90° phase shift converts the wave into circular light.

The thickness of the birefringent materials must satisfy the expression:

$$d(|n_o - n_e|) = (2m + 1) \lambda_0 / 4$$

$$m = 0, 1, 2, \dots$$

Retarders (wave plates):

Zero-order retarder (零级波片): the minimum thickness necessary to produce the required phase difference.

For example: a quartz quarter-wave plate; Birefringence: 0.0092 at 550nm => thickness: 15 μ m.

Fragile and difficult to fabricate; but large angular field-of-view.

Multiple-order retarder (多级波片): a thickness that corresponds to a whole number of 2π phase shifts plus the desired $\Delta\varphi$.

Less expensive, but sensitive to wavelength, incident angle and temperature, and narrow field-of-view.

Compound zero-order (复合零级波片): combine two multiple-order retarders whose retardance difference yields the desired value of $\Delta\varphi$.

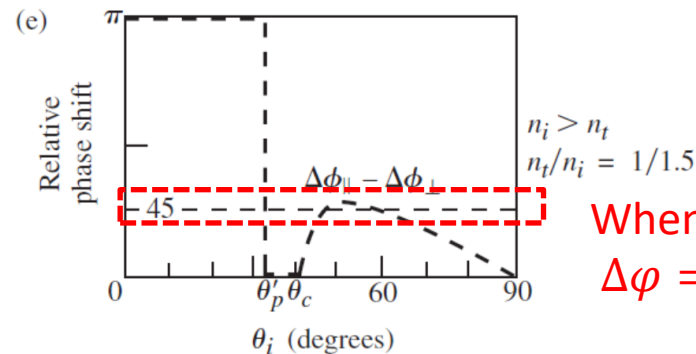
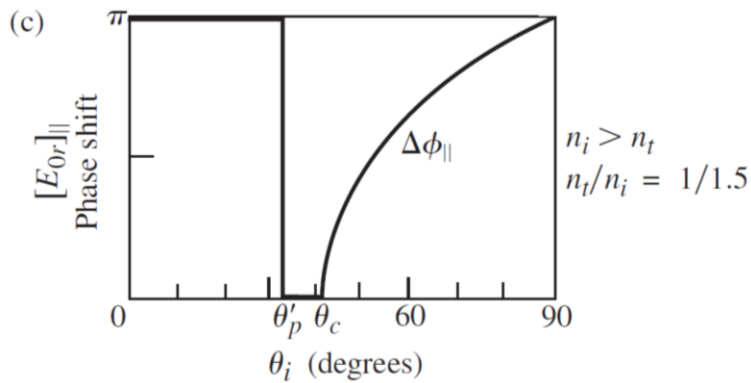
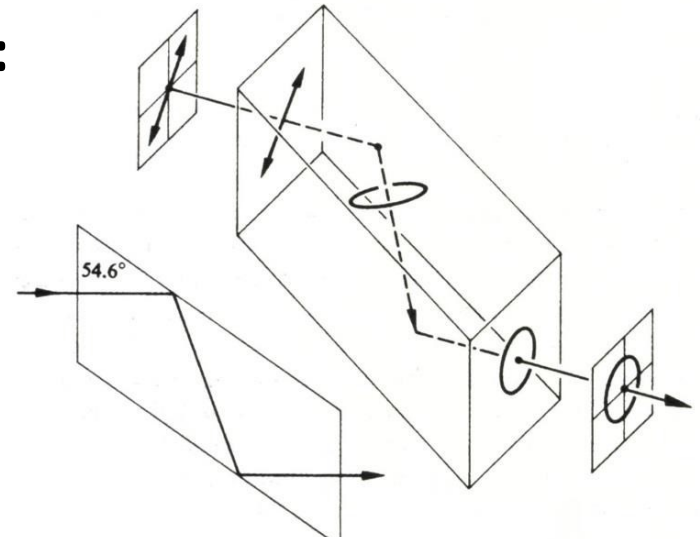
Tend to cancel the temperature variations

The Fresnel rhomb (菲涅尔斜方体) :

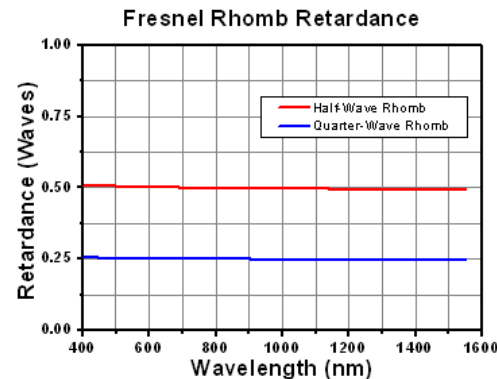
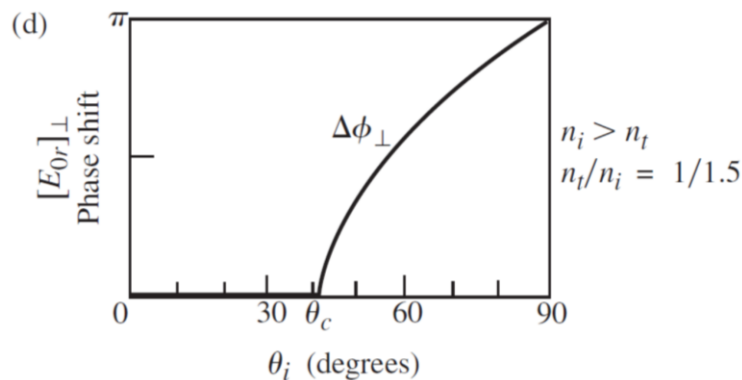
Principle: the total internal reflection introduced a relative phase difference between the two orthogonal field components

Note:

The retardance is almost independent of frequency over a larger range (全色的90°延迟器) .



When $\theta_i = 54.6^\circ$,
 $\Delta\phi = 45^\circ$

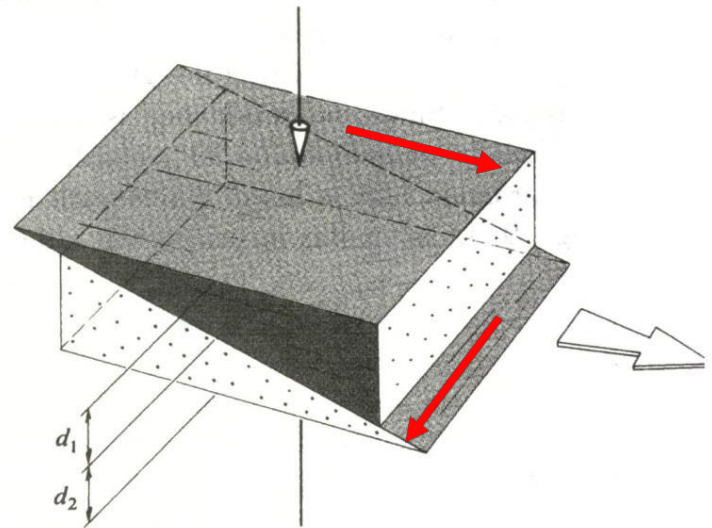


7.7.2 Compensators and variable retarders

A compensator is an optical device that is capable of impressing a controllable retardance on a wave.

Babinet compensator: the total phase difference

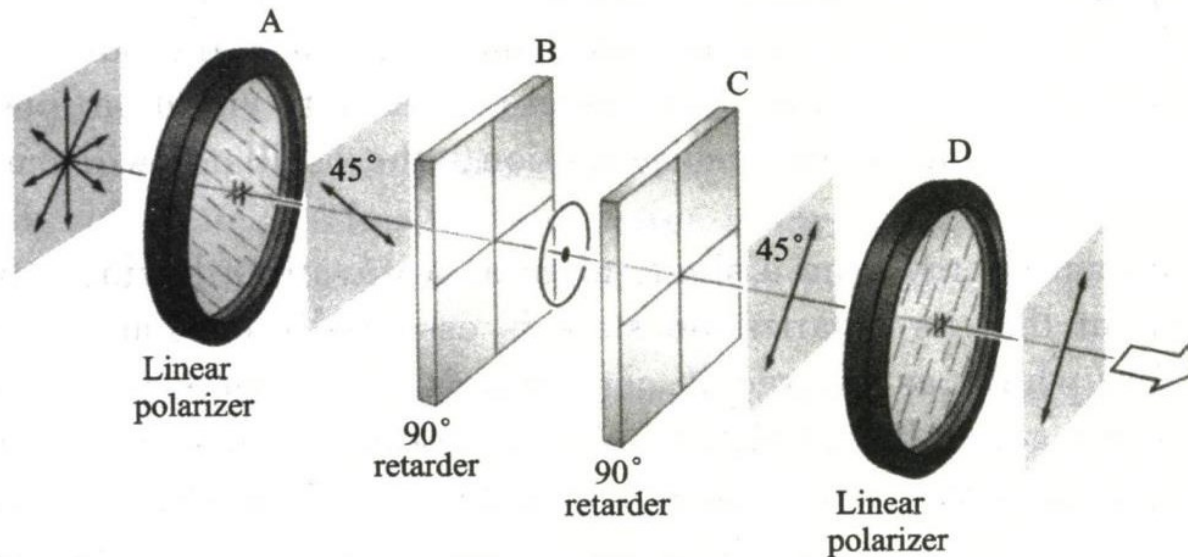
$$\Delta\varphi = \frac{2\pi}{\lambda_0} (d_1 - d_2)(|n_o - n_e|)$$



7.8 Circular Polarizers

Any series combination of an appropriately oriented linear polarizer and a 90° retarder will perform as a circular polarizer.

Example: It is used as an analyzer to determine the handedness of a wave that is already known to be circular.



- ❑ The Left-circular light entering a left-circular polarizer from the output side will be transmitted.
- ❑ The right-circular light will produce a P-state perpendicular to the transmission axis of D and so will be absorbed.

7.9 Polarization of polychromatic light

The radiant energy from a single atom:

- ❑ Wavetrain with a finite spatial extent Δl_c
- ❑ Assume its polarization state is constant within coherent time Δt_c

A typical source: a large collection of such radiating atoms, oscillating with different phases at some dominant frequency $\bar{\nu}$.

- ❑ In a very small region of the source and during very short time: the resultant wave has a polarization state.

Consider a wave whose bandwidth is very small in comparison with its mean frequency, a quasimonochromatic wave

$$\vec{E}_x(z, t) = \hat{i}E_{0x}\cos[\bar{k}z - \bar{\omega}t + \epsilon_x(t)]$$

$$\vec{E}_y(z, t) = \hat{j}E_{0y}\cos[\bar{k}z - \bar{\omega}t + \epsilon_y(t)]$$

- ❑ In a long time: the amplitude and phase will vary either independently or in some correlated fashion. So the wave is unpolarized.

7.9.2 Interference Colors

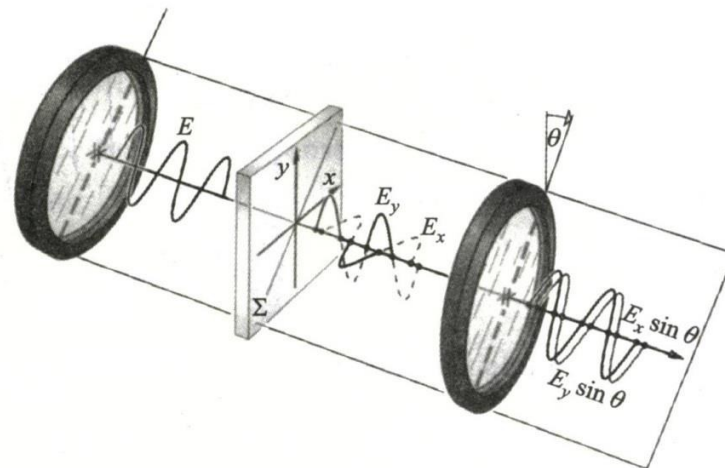
Interference colors arise from the wavelength dependence of the retardation.

$$\Delta\varphi = \frac{2\pi}{\lambda_0} d(|n_o - n_e|)$$

Example: the retardance arising some point P on retarder for blue light ($\lambda_0 = 435 \text{ nm}$) is 4π .

For incident yellow light ($\lambda_0 = 580 \text{ nm}$) is $\Delta\varphi = 3\pi$

If the blue light is parallel to the analyzer's transmission axis, the yellow light will along the extinction axis.



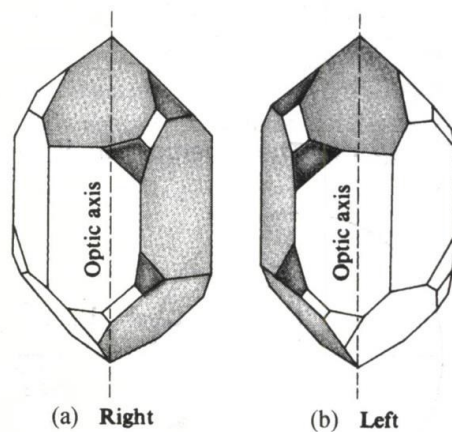
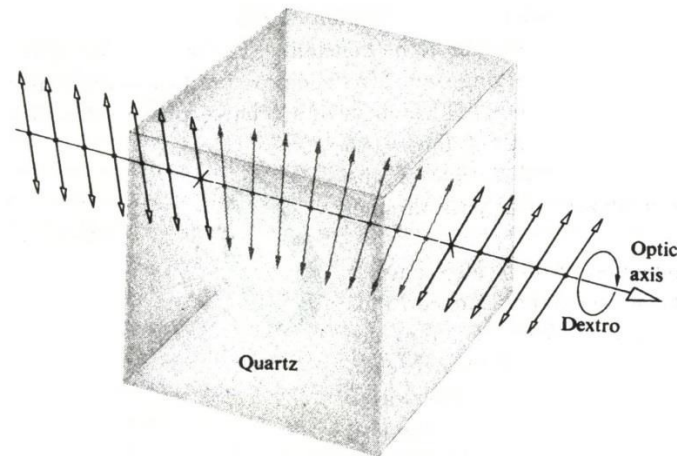
Interference colors from liquid crystals



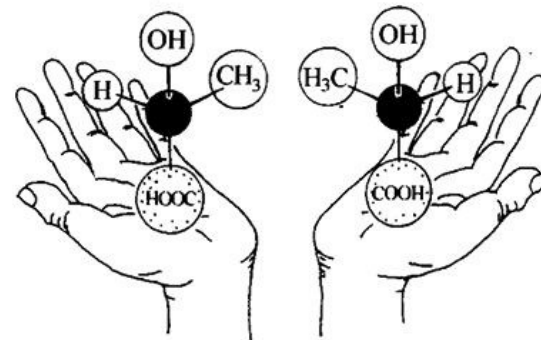
7.10 Optical Activity

Any material that causes the E-field of an incident linear plane wave to appear to rotate is said to be **optically active**.

- ❑ If the plane-of-vibration appears to have revolved clockwise: dextrorotatory or d-rotatory
- ❑ If it displaces counterclockwise: the materials is levorotatory or l-rotatory.



手性分子 chiral molecule 具有手性的分子，一般在一个C原子上按四面体方式连



接上4个互不相同的基团形成手性分子，如图示出左右两只手托出两个呈镜像关系的乳酸分子。一切具有螺旋结构的分子都是手性分子。[P]

Model: the incident linear wave is represented as a superposition of R- and L- states, and two forms of circular light propagate at different speed (circular birefringence); In traversing an optically active specimen, the two circular waves would get out-of-phase, the resultant linear wave would appear to have rotated.

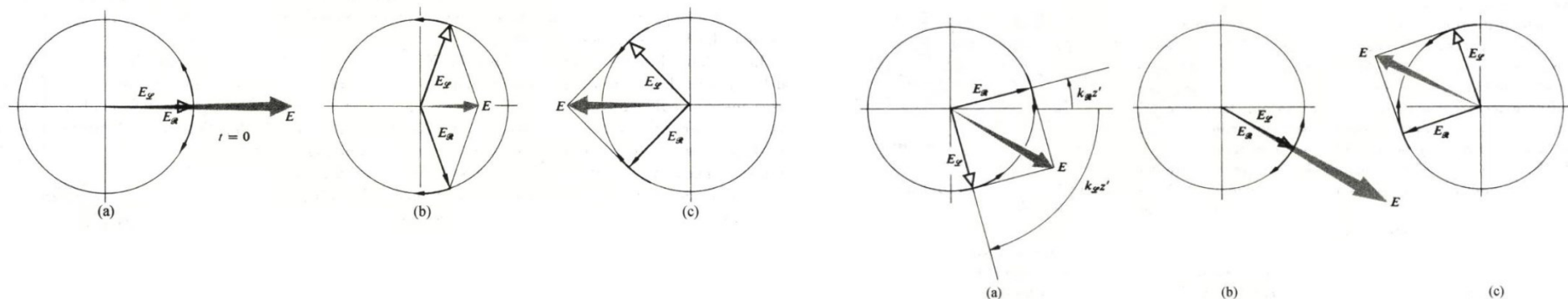
$$E_R = \frac{E_0}{2} [\hat{i} \cos(k_R z - \omega t) + \hat{j} \sin(k_R z - \omega t)]$$

$$E_L = \frac{E_0}{2} [\hat{i} \cos(k_L z - \omega t) - \hat{j} \sin(k_L z - \omega t)]$$

The resultant disturbance:

$$E = E_0 \cos[(k_R + k_L)z/2 - \omega t] \left[\hat{i} \cos \frac{(k_R - k_L)z}{2} + \hat{j} \sin \frac{(k_R - k_L)z}{2} \right]$$

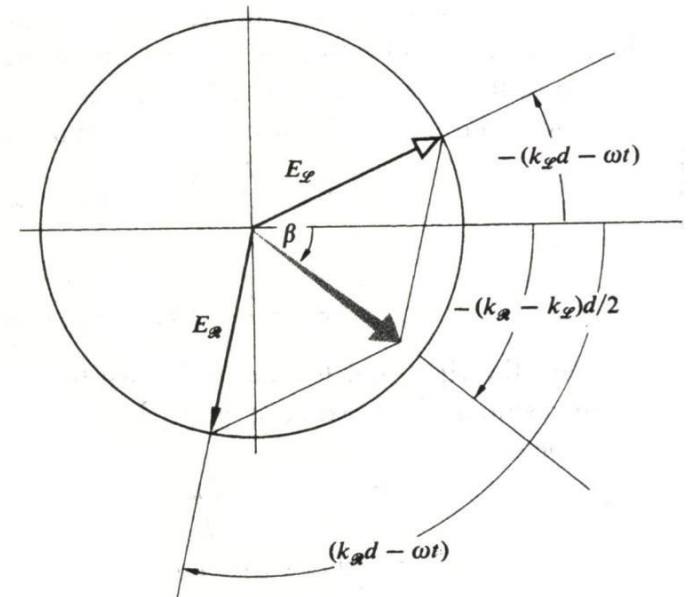
The superposition of a R- and an L- state at $z=0$ The superposition of a R- and an L- state at $z=z'$



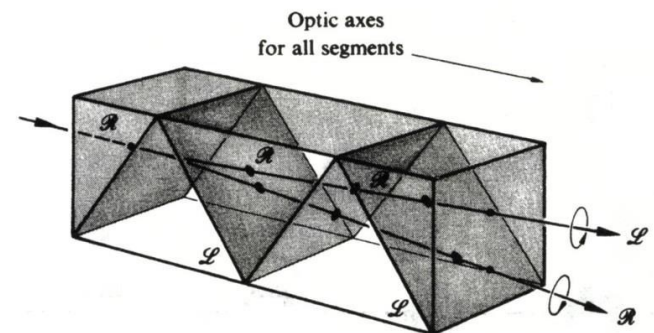
At point z , there is an angle $\beta = -\frac{(k_R - k_L)z}{2}$ with respect to its original orientation.

If the medium has a thickness d , the angle through which the plane-of-vibration rotates is:

$$\beta = \frac{\pi d}{\lambda_0} (n_L - n_R)$$



Separate the constituent R- and L- states of a linear beam using the composite prism



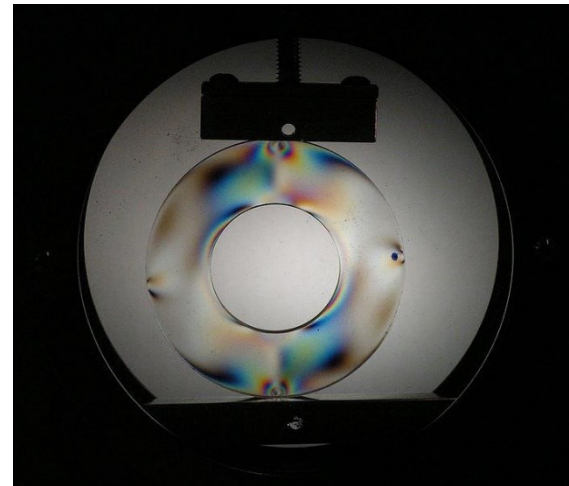
7.11 Induced Optical Effect-Optical Modulators

I. Photoelasticity (mechanical birefringence or stress birefringence)

- ❑ Normally transparent isotropic substance could be made optically anisotropic by the application of mechanical stress.
- ❑ The effective optic axis is in the direction of the stress and the induced birefringence is proportional to the stress.

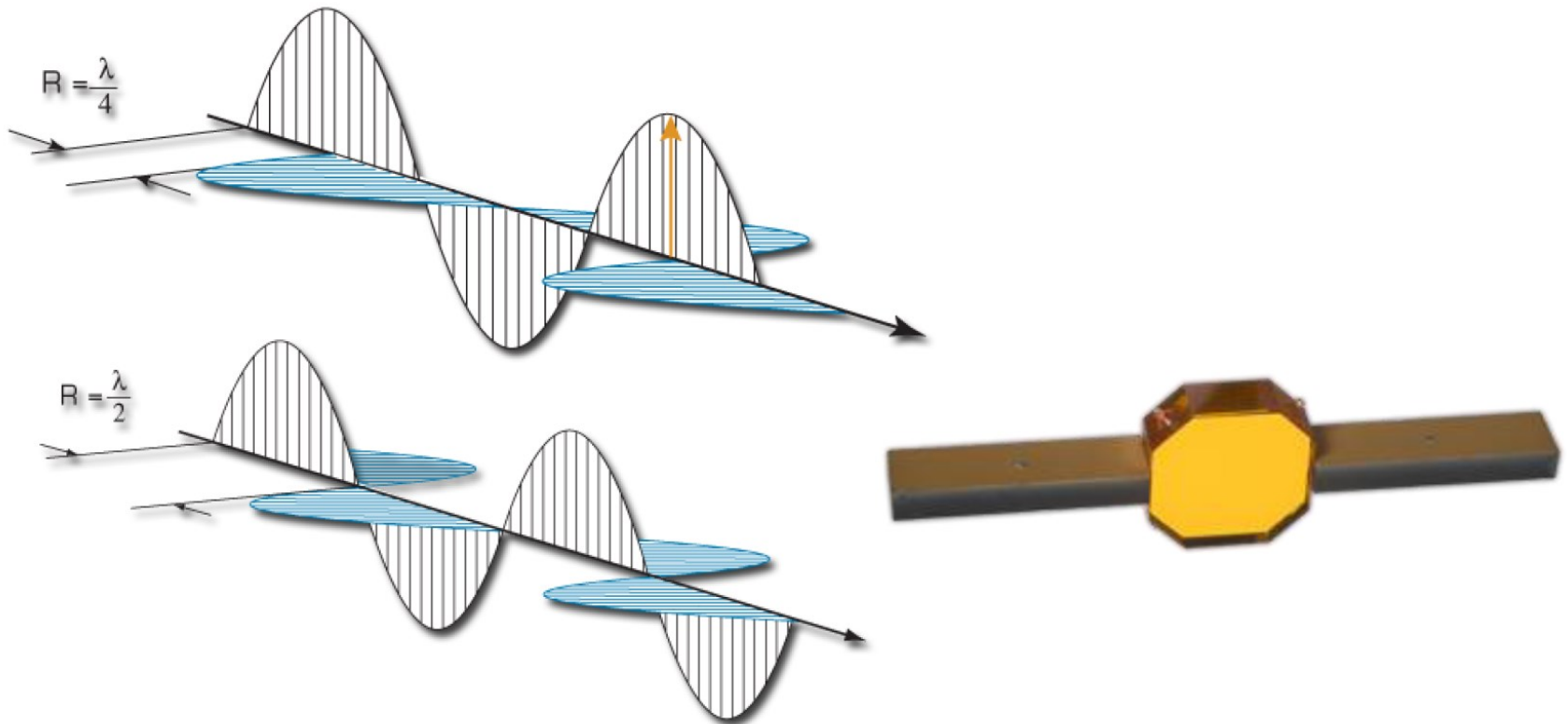
Example: a plate under vertical tension.

- ❑ σ_1 : the maximum principle stress in the vertical direction.
- ❑ σ_2 : the minimum principal stress horizontally. It is zero.
- ❑ The retardance $\propto$ the principal stress difference ($\sigma_1 - \sigma_2$)



Photoelastic Modulation

Principle : a sample of transparent solid material is stressed by compression or stretching, the material becomes birefringent, that is, different linear polarizations of light have slightly different speeds of light when passing through the material.



II. The Faraday Effect

Phenomenon: the plane-of-vibration of linear light incident on a piece of glass rotated when a strong magnetic field was applied in the propagation direction. (**magneto-optic effect**)

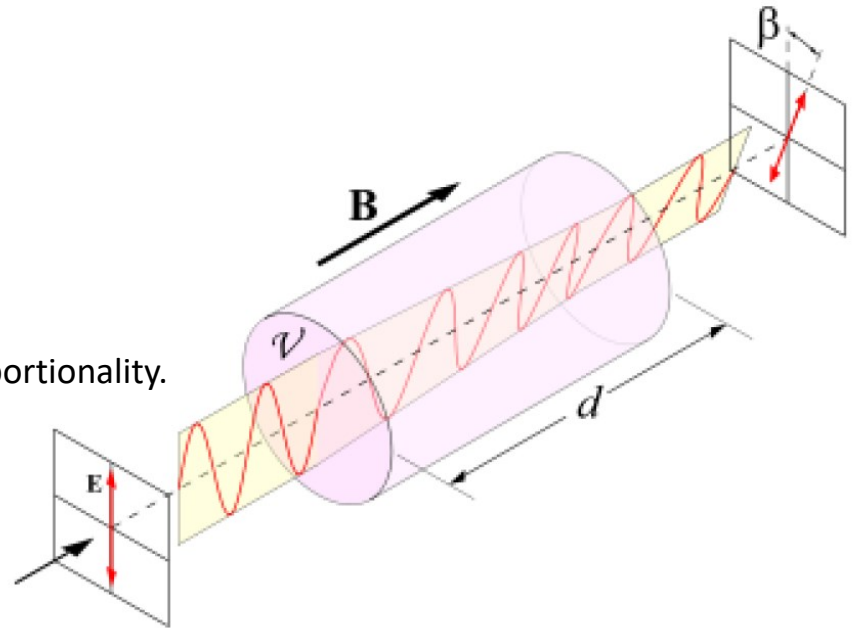
Empirical expression:

$$\beta = VBd$$

B: the static magnetic flux density;

d: the length of medium traversed;

V [Verdet constant (费尔德常数)]: a factor of proportionality.



Positive V: l-rotatory when the light moves parallel to B; d-rotatory when it propagates antiparallel to B.

Note: no such reversal of handedness occurs in the case of natural optical activity.

Classical argument for nonmagnetic materials:

Consider the incident light to be circular and monochromatic.

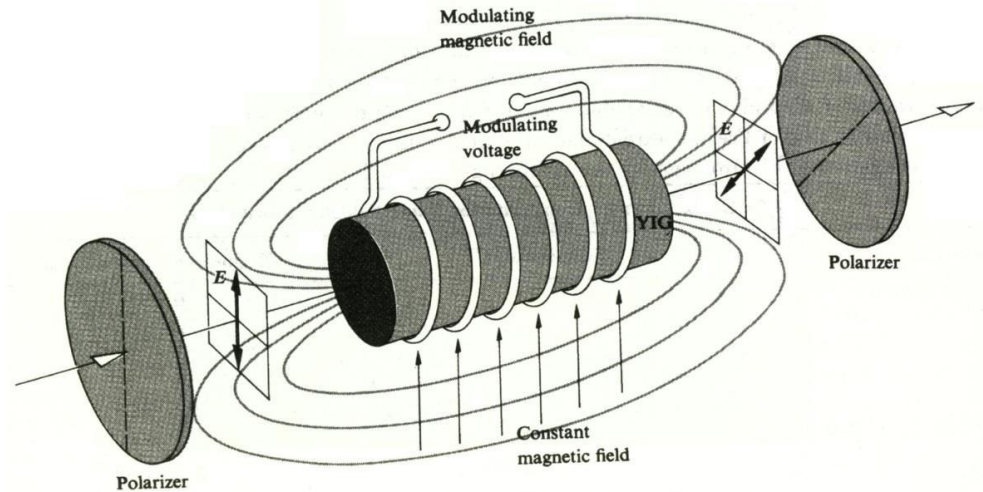
- ❑ An elastically bound electron will take on a steady-state circular orbit being driven by the rotating E-field of the wave.
- ❑ The introduction of a magnetic field perpendicular to the plane of the orbit will result in a radial force F_m on the electron.
- ❑ Its direction is determined by the handedness of the light and the direction of the B-field.
- ❑ The total radial force has two values, so two radii of the orbit.

For a given magnetic field, there will be two possible values of the electric dipole moment, the polarization, the permittivity, as well two values of the index of refraction, n_R and n_L .

Application of Faraday Effect

Optical modulators

Materials: synthetic magnetic crystal yttrium-iron garnet (YIG, 钇铁石榴石), which is used in infrared light.



Other magneto-optic effects

Voigt (in vapors) and Cotton-Mouton (in liquids) Effects: both arise when a constant magnetic field is applied to a transparent medium perpendicular to the direction of propagation of the incident light beam.

Principle: medium displays birefringence similar to that of a uniaxial crystal whose optic axis is in the direction of the dc magnetic field.

The difference of indices of refraction is proportional to the square of the applied magnetic field.

III. The Kerr Effect

Phenomenon: an isotropic transparent substance becomes birefringent when placed in an electric field E . (electro-optic effect)

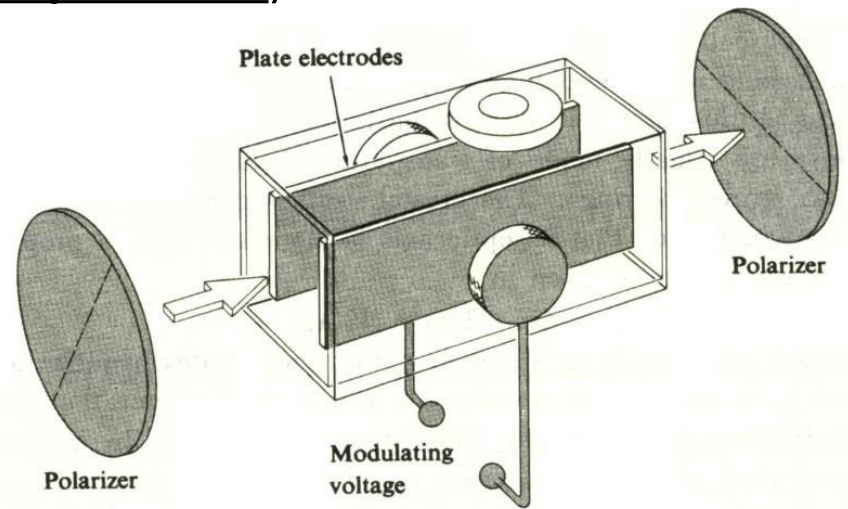
Reason: a partial alignment of anisotropic molecules by the E -field in liquids

The medium takes on the characteristics of a uniaxial crystal whose optic axis corresponds to the direction of the applied field.

The difference of two indices (parallel and perpendicular to the E -field) $\Delta n = n_{\parallel} - n_{\perp}$:

$$\Delta n = \lambda_0 K E^2$$

K : the Kerr constant.



Kerr shutter or optical modulator:

The retardation

$$\Delta\phi = 2\pi K l V^2 / d^2$$

l : the effective length;

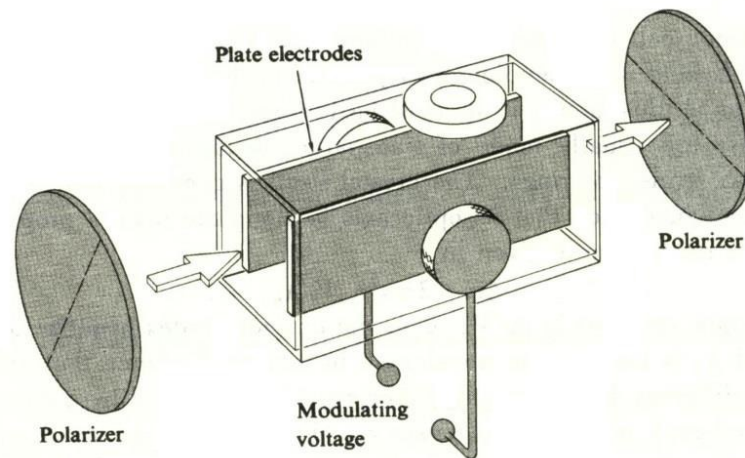
d : the distance of separation

For example: a nitrobenzene cell with $d=1\text{cm}$, l is several cm.

A half-wave plate: $V=3\times 10^4\text{V}$

Respond frequency: up to 10^{10}Hz

Application: shutters in high-speed photography; light-beam choppers



Homework

□ 8.45, 8.46, 8.49